B. Tech (Information Technology & Mathematical Innovations)

Semester-VII (Revised)

For the session 2026,2027,2028

S. No.	Course No.	Course Title	Credits	Internal Evaluation	External Evaluation	Hours/Week			Total Marks
						L	P/EL	T	
1	UMJDTT-711	Computer and Brain: Knowledge Discovery and Artificial Intelligence with Python / MATLAB	4	40	60	3	1	0	100
2		**Elective**	4	40	60	3	1	0	100
	UMJDTT-712	Quantum Computing for beginners							
	UMJDTT-713	Emerging Technologies							
	UMJDTT-714	Crpto Currency & Block Chain Technologies							
3	UMJDTT-715	Robotics Process Automation & Drones	4	40	60	3	1	0	100
4	UMJDTT-716	AI in Neuro & Cognitive Computing	4	40	60	3	1	0	100
5	UMJDTT-717	Numerical methods: from approximation to perfection	4	40	60	3	1	0	100
6	UMJDTP-718**	Project 1	6	100	50	0	6	0	150
7	UMJDTP-719 **	Project 2	6	100	50	0	6	0	150

***L- Lecture, *T- Tutorial, *P/EL- Practical / Experiential Learning**
**** Guidelines attached**

B. Tech (Information Technology & Mathematical Innovations)

Semester-VIII (Revised)

For the session 2026,2027,2028

S. No.	Course No.	Course Title	Credits	Internal Evaluation	External Evaluation	Hours/Week			Total Marks
						L	P/EL	T	
1	UMJDTT-811	Advance, Generative AI & Agents	4	40	60	3	1	0	100
2	UMJDTT-812	The Science of Chance	4	40	60	3	1	0	100
3	UMJDTT-813	Computer Vision	4	40	60	3	1	0	100
4	UMJDTT-814	IoT Sensor Networks & Data Analytics	4	40	60	3	1	0	100
5	UMJDTT-815	AI Driven Digital Forensics and Cyber threat Intelligence	4	40	60	3	1	0	100
6	UMJDTP-816**	Project 1	6	100	50	0	6	0	150
7	UMJDTP-817**	Project 2	6	100	50	0	6	0	150

***L- Lecture, *T- Tutorial, *P/EL- Practical / Experiential Learning**

**** Guidelines attached**

UNIVERSITY OF JAMMU
Four Year Innovative Undergraduate Program
(Design Your Degree)
B. Tech (Information Technology & Mathematical Innovations) (Revised)
Semester VII
(For the session 2026, 2027, 2028)

Course Code: UMJDTT-711

Course Title: Computer and Brain: Knowledge Discovery and Artificial Intelligence with Python / MATLAB

Credits: 04
Contact Hours: 60

Maximum Marks : 100
Internal Evaluation : 40
External Evaluation : 60

COURSE DESCRIPTION

This course introduces the fundamentals of Machine Learning and Knowledge Discovery through practical implementation using Python or MATLAB. Students will learn how to prepare data, extract useful features, build predictive models, discover hidden patterns, evaluate machine learning solutions using real-world datasets and intelligent decision-making systems. The course emphasizes practical implementation of AI algorithms using Python and MATLAB for data analysis, pattern recognition, predictive modeling, image processing, and intelligent applications. By the end of the course, students will gain hands-on experience in building AI models, discovering knowledge from real-world datasets, and developing brain-inspired intelligent systems while understanding ethical and responsible AI practices.

LEARNING OUTCOMES

By the end of this course, students will be able to:

1. Evaluate machine learning models using appropriate performance metrics.
2. Develop practical machine learning solutions using Python or MATLAB while understanding responsible and ethical AI practices.Apply machine learning and data mining techniques to discover meaningful patterns from real-world datasets.
3. Develop intelligent applications using Python or MATLAB for classification, prediction, image analysis, and decision support.
4. Evaluate AI models using appropriate performance measures while applying ethical and responsible AI practices.

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Credits: 04
Contact Hours: 60
Maximum Marks : 100
Internal Evaluation : 40
External Evaluation : 60

MODULE I : Understanding Real-World Problems through AI and Data

Understanding how Artificial Intelligence transforms healthcare, finance, agriculture, education, manufacturing and smart cities. Exploring structured, unstructured, image, text, audio and business data for intelligent decision making. Discovering meaningful patterns and insights from real-world datasets using Python/MATLAB. Visualizing data to support business, scientific and engineering decisions.

Hands-On Activities

1. Exploratory Analysis of Road Accident Data: Identify accident trends using visualization and descriptive statistics.
2. Air Quality Monitoring Dashboard: Analyze pollution data and create interactive visualizations.
3. Retail Sales Data Exploration: Discover seasonal sales patterns and customer purchasing behavior.

MODULE II: Developing Intelligent Prediction and Decision-Making Systems

Building predictive systems using historical data for forecasting and trend analysis. Developing classification models for healthcare, finance and cybersecurity applications. Discovering customer groups and hidden patterns using clustering techniques. Preparing real-world datasets through preprocessing, feature engineering and model evaluation. Improving prediction accuracy through feature selection, validation and performance analysis.

Hands-On Activities

1. Preprocess Electronic Health Records : Handle missing values, encode patient information, and normalize clinical features.
2. Loan Application Data Preparation : Build a complete preprocessing pipeline for banking datasets.
3. Feature Engineering for Employee Attrition Prediction : Select important features influencing employee turnover.

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4. Energy Consumption Data Cleaning : Prepare smart meter datasets for predictive analytics using Scikit-learn pipelines.

MODULE III: Supervised Machine Learning for Intelligent Prediction Systems

Developing disease diagnosis and clinical decision support systems using Logistic Regression, Decision Trees and Random Forest. Building house price, energy consumption and sales forecasting applications using Linear Regression. Designing financial risk assessment and loan approval systems using Decision Trees, Support Vector Machines (SVM) and Naive Bayes. Evaluating prediction models using Accuracy, Precision, Recall, F1-Score, Confusion Matrix, MAE and Cross Validation.

Hands-On Activities

1. Credit Risk Assessment : Predict whether a loan applicant is likely to default.
2. Heart Disease Risk Prediction : Compare Logistic Regression, Decision Tree, KNN, and SVM.
3. Crop Disease Classification : Classify crop health conditions using supervised learning algorithms.

MODULE IV: Unsupervised Machine Learning for Knowledge Discovery

Developing customer segmentation and personalized marketing solutions using K-Means Clustering. Building user profiling and recommendation systems using Hierarchical Clustering. Designing fraud detection and anomaly detection applications using outlier detection techniques. Discovering shopping patterns and product associations using Association Rule Mining (Apriori). Visualizing high-dimensional healthcare, IoT and business datasets using Principal Component Analysis (PCA). Extracting hidden knowledge from large datasets using clustering, dimensionality reduction and pattern discovery techniques.

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Credits: 04
Contact Hours: 60
Maximum Marks : 100
Internal Evaluation : 40
External Evaluation : 60

Hands-On Activities

1. Customer Segmentation for E-commerce : Group customers based on purchasing behavior using K-Means.
2. Telecommunication Customer Profiling : Discover user groups for targeted marketing.
3. Retail Market Basket Analysis : Identify frequently purchased product combinations using association rules.
4. Credit Card Fraud Detection : Detect unusual transaction patterns using anomaly detection techniques.
5. Dimensionality Reduction of Sensor Data : Apply PCA to visualize high-dimensional IoT datasets.

MODULE V: Reinforcement Learning and Intelligent AI Systems

Developing autonomous navigation systems using Agent, Environment, State, Action, Reward and Q-Learning. Building intelligent game-playing and decision-making systems using Reinforcement Learning. Creating adaptive AI systems for autonomous vehicles and smart automation using Reinforcement Learning concepts.

Hands-On Activities

1. Autonomous Robot Navigation in Grid World : Train a Q-learning agent to reach a target while avoiding obstacles.
2. Warehouse Path Optimization using FrozenLake : Learn optimal navigation policies through reinforcement learning.
3. Comparison of Machine Learning, Reinforcement Learning, and Deep Learning Applications : Case study and implementation discussion.

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Course Title: Computer and Brain: Knowledge Discovery and Artificial Intelligence with Python / MATLAB

Credits: 04
Contact Hours: 60

Maximum Marks : 100
Internal Evaluation : 40
External Evaluation : 60

MINOR PROJECT FOR THIS SEMESTER

1. AI-Based Public Health Surveillance and Disease Outbreak Prediction System
2. Precision Agriculture Decision Support and Crop Intelligence System
3. AI-Powered Smart Energy Consumption Forecasting Platform
4. Intelligent Waste Management and Route Optimization System
5. Retail Demand Forecasting and Inventory Optimization System
6. Road Infrastructure Damage Detection and Maintenance Planning System

PEDAGOGY

The course will be delivered through an application-oriented and experiential learning approach emphasizing practical implementation using Python and MATLAB. Students will learn by analyzing real-world datasets, developing machine learning models, building intelligent applications, and solving practical problems using AI techniques. The course encourages students to explore knowledge discovery techniques and brain-inspired AI through real-world applications while promoting ethical and responsible use of Artificial Intelligence.

GUIDELINES FOR ASSESSMENT AND EVALUATION FOR SEMESTERS V - VIII OF FYUGP (DYD)

1. Assessment and Evaluation Model

Features of this Model of Assessment and Evaluation are:

- Domain knowledge: Tested through class tests and end-sem examination.
- Creativity: Assignments, seminars, open-ended examination questions.
- Hands-on experience: Practical tasks and project work to build real-world problem-solving skills.

Following scheme of assessment and evaluation shall be followed for the last two years of FYUGP (DYD):

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Semester VII
(For the session 2026, 2027, 2028)

Course Code: UMJDTT-711
Course Title: Computer and Brain: Knowledge Discovery and Artificial Intelligence with Python / MATLAB
Credits: 04
Contact Hours: 60
Maximum Marks : 100
Internal Evaluation : 40
External Evaluation : 60

1.1 Continuous Internal Assessment (CIA) - 40 Marks

This ensures sustained engagement and assessment beyond the final exam. CIA shall have the following components:

- Class Tests and Quizzes (10 Marks): This shall focus on concepts clarity, analytical ability comprising 2 or 3 short written or online tests.
- Assignments and Creative Tasks (10 Marks): This shall be focused on Problem-solving or reflective assignments and Open-ended questions to assess creativity.
- Practical and Hands-on Activities (15 Marks): This shall have the activities like Lab work, field tasks, coding assignments, design exercises, case study etc.
- Seminar and Presentation (5 Marks): This shall have individual or group presentation to be evaluated for communication, originality, creativity, and depth.

1.2 End-Semester Examination - 60 Marks

This is designed to test comprehensive domain knowledge and applications. The question paper shall have three sections:

- Section A - 20 Marks: Short answer questions to test fundamentals and breadth of domain knowledge.
- Section B - 20 Marks: Problem-solving and applications of the domain knowledge. The problems need to be conceptual numerical and case-based related to real world applications.
- Section C - 20 Marks: This is to be devoted to essay/analytical questions (choice-based) and have to be Open-ended, requiring synthesis, critical thinking, and creativity.

RECOMMENDED TEXTBOOKS

1. Artificial Intelligence: A Modern Approach — Stuart Russell and Peter Norvig, Pearson.
2. Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow — Aurélien Géron, O'Reilly Media.
3. Aurélien Géron, *Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow*, 3rd Edition, O'Reilly Media, 2022.
4. Ethem Alpaydin, *Introduction to Machine Learning*, 4th Edition, MIT Press, 2020.
   Christopher M. Bishop and Hugh Bishop, *Deep Learning: Foundations and Concepts*, Springer, 2024.

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B. Tech (Information Technology & Mathematical Innovations) (Revised)
Semester VII
(For the session 2026, 2027, 2028)

Course Code: UMJDTT-712	Course Title: Quantum Computing: Concepts, Applications, and Future Technologies
Credits: 04	Maximum Marks : 100
Contact Hours: 60	Internal Evaluation : 40
	External Evaluation : 60

COURSE DESCRIPTION

This course introduces students to the emerging field of Quantum Technologies integrated with Artificial Intelligence (AI), providing a foundation in quantum computing, quantum communication, quantum sensing, and quantum materials. Students learn the principles of qubits, superposition, entanglement, quantum gates, and hybrid quantum-classical computing while exploring how AI techniques enhance the design, optimization, security, and performance of quantum systems. Through hands-on programming using Python, Qiskit, PennyLane, and IBM Quantum platforms, learners gain practical experience in building quantum circuits, quantum machine learning models, quantum neural networks, AI-assisted quantum communication systems, and intelligent quantum sensing applications. The course also introduces AI-driven material discovery for next-generation quantum hardware. Emphasis is placed on experiential learning through laboratory exercises, mini-projects, and real-world case studies, enabling students to develop the knowledge and practical skills required for future research and innovation in quantum technologies and intelligent computing.

LEARNING OUTCOMES

By the end of this course, students will be able to:

1. Write basic code to simulate quantum systems using Python tools like Qiskit and
2. PennyLane.
3. Train a Quantum Neural Network to recognize patterns in standard datasets.
4. Build an AI security program that stops eavesdroppers on a simulated quantum network.
5. Use AI to predict the best physical materials for building future quantum chips.

MODULE I: Foundations of Quantum Computing

Evolution of computing: From Classical to Quantum, Why Quantum Computing?, Limitations of Classical Computing, Global Developments and the Future of Quantum Technologies, Current Challenges, Limitations, and Future Prospects of Quantum Technologies,

Hands-on Activities

1. Setting up Python, Qiskit, and basic machine learning libraries.
2. Coding a simple quantum coin flip (superposition) and using basic AI code to chart the probability of measurement outcomes.

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Credits: 04	Maximum Marks : 100
Contact Hours: 60	Internal Evaluation : 40
	External Evaluation : 60

MODULE II: Fundamentals of Quantum Computing

Classical Bits vs. Quantum Bits (Qubits), Principles of Superposition, Concept of Entanglement Quantum Interference and Quantum Measurement (Conceptual Understanding)

Hands-on Activites

1. Connecting different quantum gates in Qiskit to observe how data changes step by step.
2. Replacing a standard AI data layer with a quantum circuit using PennyLane.
3. Training the combined Quantum-AI network to accurately sort a simple dataset of flower
4. measurements.

MODULE III: Quantum Computers and Ecosystem

Architecture of Quantum Computers (Overview), Types of Qubits and Quantum Hardware, Quantum Gates and Quantum Circuits (Introduction), Major Quantum Computing Platforms and Cloud-Based Quantum Computing.

Hands-on Activities

1. Writing code to simulate two computers sharing a secret quantum password.
2. Adding a simulated hacker to the code to observe how the quantum data gets corrupted.
3. Building an AI alarm system that analyzes the data errors and alerts the user when a hacker is present.

MODULE IV: Applications of Quantum Computing

Quantum Computing in Artificial Intelligence and Machine Learning, Cryptography and Cybersecurity, Healthcare, Drug Discovery, and Life Sciences, Optimization, Finance, Logistics, Climate Science, and Smart Systems.

Hands-on Activities

1. Generating a set of fake data that mimics a highly sensitive but noisy quantum compass.
2. Processing the data using classical math filters to understand why traditional methods fail.
3. Training an AI deep learning model to successfully strip away the static and recover the perfect signal.

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Credits: 04	Maximum Marks : 100
Contact Hours: 60	Internal Evaluation : 40
	External Evaluation : 60

MODULE V: Future of Quantum Technologies and Its Verticals

India's National Quantum Mission and Global Quantum Initiatives, Emerging Research Trends, Career Opportunities, and Ethical Considerations in Quantum Technologies.

Quantum Computing: Solving complex computational problems and enabling next-generation computing

Quantum Communication: Secure communication, quantum networks, and the Quantum Internet

Quantum Sensing & Metrology: High-precision sensing, measurement, navigation, and imaging technologies

Quantum Materials & Devices: Superconducting circuits, trapped ions, photonic systems, quantum dots, and other enabling quantum hardware

Hands-on Activities

1. Exploring a public database of chemicals and materials used in advanced electronics.
2. Turning complex chemical structures into simple data graphs that an AI can read.
3. Training an AI to predict which new material combination would make the most stable quantum chip.

Minor Projects for this semester

1. Compare a classical AI to a Quantum-AI to see which one is faster at predicting stock market trends based on a provided dataset.
2. Build a simulated quantum chat network and train an AI to automatically reroute messages if it detects environmental noise or a hacker.
3. Feed an AI a list of 1,000 different metal alloys and have it predict which one would keep a quantum chip stable for the longest amount of time.

Recommended Software Tools

Python, Jupyter Notebook, IBM Quantum Lab, IBM Quantum Composer, Qiskit, Cirq,PennyLane, Google Colab, VS Code, Git & GitHub, ChatGPT / GitHub Copilot / Google Gemini/ Claude by Anthropic, TensorFlow Quantum

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Course Code: UMJDTT-712	Course Title: Quantum Computing: Concepts, Applications, and Future Technologies
Credits: 04	Maximum Marks : 100
Contact Hours: 60	Internal Evaluation : 40
	External Evaluation : 60

PEDAGOGY

The course will be delivered through a blended and experiential learning approach, combining interactive lectures, laboratory sessions, hands-on programming, case studies, and project-based learning. Students will gain practical experience using Python, Qiskit, PennyLane, and IBM Quantum tools to implement quantum and AI applications. Learning will be reinforced through coding exercises, discussions, assignments, and mini projects, with continuous assessment to promote conceptual understanding, problem-solving, and research-oriented skills.

GUIDELINES FOR ASSESSMENT AND EVALUATION FOR SEMESTERS V - VIII OF FYUGP (DYD)

1. Assessment and Evaluation Model

Features of this Model of Assessment and Evaluation are:

- Domain knowledge: Tested through class tests and end-sem examination.
- Creativity: Assignments, seminars, open-ended examination questions.
- Hands-on experience: Practical tasks and project work to build real-world problem-solving skills.

Following scheme of assessment and evaluation shall be followed for the last two years of FYUGP (DYD):

1.1 Continuous Internal Assessment (CIA) - 40 Marks

This ensures sustained engagement and assessment beyond the final exam. CIA shall have the following components:

- Class Tests and Quizzes (10 Marks): This shall focus on concepts clarity, analytical ability comprising 2 or 3 short written or online tests.
- Assignments and Creative Tasks (10 Marks): This shall be focused on Problem-solving or reflective assignments and Open-ended questions to assess creativity.

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Semester VII
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Course Code: UMJDTT-712	Course Title: Quantum Computing: Concepts, Applications, and Future Technologies
Credits: 04	Maximum Marks : 100
Contact Hours: 60	Internal Evaluation : 40
	External Evaluation : 60

- Practical and Hands-on Activities (15 Marks): This shall have the activities like Lab work, field tasks, coding assignments, design exercises, case study etc.
- Seminar and Presentation (5 Marks): This shall have individual or group presentation to be evaluated for communication, originality, creativity, and depth.

1.2 End-Semester Examination - 60 Marks

This is designed to test comprehensive domain knowledge and applications. The question paper shall have three sections:

- Section A - 20 Marks: Short answer questions to test fundamentals and breadth of domain knowledge.
- Section B - 20 Marks: Problem-solving and applications of the domain knowledge. The problems need to be conceptual numerical and case-based related to real world applications.
- Section C - 20 Marks: This is to be devoted to essay/analytical questions (choice-based) and have to be Open-ended, requiring synthesis, critical thinking, and creativity.

Recommended Textbooks

1. Michael A. Nielsen and Isaac L. Chuang, Quantum Computation and Quantum Information, Cambridge University Press.
2. Eleanor Rieffel and Wolfgang Polak, Quantum Computing: A Gentle Introduction, MIT Press.
3. Jack D. Hidary, Quantum Computing: An Applied Approach, Springer.
4. N. David Mermin, Quantum Computer Science: An Introduction, Cambridge University Press.
5. Phillip Kaye, Raymond Laflamme, and Michele Mosca, An Introduction to Quantum Computing, Oxford University Press.
6. Chris Bernhardt, Quantum Computing for Everyone, MIT Press.
7. Eric R. Johnston, Nic Harrigan, and Mercedes Gimeno-Segovia, Programming Quantum Computers, O'Reilly.
8. Scott Aaronson, Quantum Computing Since Democritus, Cambridge University Press.

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Semester VII
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Course Code: UMJDTT-713
Credits: 04
Contact Hours: 60

Course Title: Emerging Technologies
Maximum Marks : 100
Internal Evaluation : 40
External Evaluation : 60

COURSE DESCRIPTION

This course provides an application-oriented overview of emerging digital technologies that are driving innovation across modern industries. Students will explore how Artificial Intelligence (AI), Internet of Things (IoT), Cloud Computing, Edge Computing, Blockchain, Big Data, Generative AI, Augmented Reality/Virtual Reality (AR/VR), and Robotic Process Automation (RPA) are transforming healthcare, manufacturing, agriculture, finance, education, smart cities and sustainable development. The course enables students to understand emerging technologies through real-world applications, hands-on activities, case studies and mini-projects. By the end of the course, students will be able to identify suitable technologies for solving engineering problems and develop intelligent, secure and sustainable digital solutions.

LEARNING OUTCOMES

By the end of this course, students will be able to:

1. Explain the principles and real-world applications of emerging technologies, including AI, IoT, Blockchain, Cloud Computing, AR/VR and RPA.
2. Develop simple intelligent applications using modern digital technologies, cloud platforms and AI-based tools to address engineering and societal challenges.
3. Analyze real-world datasets and connected systems to support data-driven decision making using Big Data analytics, Machine Learning and IoT.
4. Design secure, intelligent and connected digital solutions by integrating multiple emerging technologies for industrial and business applications.
5. Evaluate emerging technology solutions considering ethical, privacy, security, sustainability and future technological trends.

Module I: Foundations of Next-Generation Digital Technologies

Understanding how Extended Reality (XR), Digital Twins, 6G, Advanced Semiconductor Technologies and Additive Manufacturing are transforming healthcare, manufacturing, aerospace, education, entertainment and smart cities. Exploring Industry 5.0 and intelligent cyber-physical systems. Understanding digital transformation through immersive, connected and intelligent technologies. Identifying engineering problems suitable for next-generation technologies. Developing innovation workflows for future digital enterprises.

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Course Code: UMJDTT-713
Credits: 04
Contact Hours: 60

Course Title: Emerging Technologies
Maximum Marks : 100
Internal Evaluation : 40
External Evaluation : 60

Hands-On Activities

1. Analyze case studies on Industry 5.0 and smart manufacturing.
2. Compare XR, Digital Twins, AI and IoT applications for solving engineering problems.
3. Prepare a technology roadmap for a smart factory or smart hospital.
4. Present emerging technology applications in different industries.

Module II: Learning Immersive and Intelligent Systems

Developing immersive learning and training applications using Augmented Reality (AR), Virtual Reality (VR) and Mixed Reality (MR). Building spatial computing applications for engineering visualization and remote collaboration. Designing virtual simulations for education, healthcare and industrial training. Creating interactive digital environments for human-computer interaction. Understanding user experience and interaction in immersive systems.

Hands-On Activities

1. Create a simple AR application using Unity or an AR platform.
2. Develop a virtual tour of a campus or laboratory.
3. Build an interactive VR safety training simulation.
4. Design a basic mixed reality engineering visualization.

Module III: Building Connected Digital Twins and Intelligent Infrastructure

Developing digital twin solutions for manufacturing, healthcare, buildings and smart cities. Building predictive maintenance systems using Digital Twins, IoT and sensor data. Designing next-generation communication systems using 6G, satellite communication and Non-Terrestrial Networks (NTN). Developing real-time monitoring and simulation platforms for connected infrastructure. Integrating cloud computing and edge technologies for intelligent digital systems.

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Credits: 04
Contact Hours: 60

Course Title: Emerging Technologies
Maximum Marks : 100
Internal Evaluation : 40
External Evaluation : 60

Hands-On Activities

1. Design a digital twin of a smart factory or smart building.
2. Simulate predictive maintenance using sensor data.
3. Compare 5G and 6G use cases for smart cities.
4. Develop a connected monitoring dashboard.

Module IV: Developing Intelligent Hardware and Smart Manufacturing Solutions

Building AI-enabled edge devices using next-generation semiconductor technologies. Developing energy-efficient intelligent systems using neuromorphic computing concepts. Designing rapid prototyping and product development workflows using 3D printing. Creating adaptive manufacturing solutions using 4D printing and smart materials. Integrating intelligent hardware with AI, IoT and robotics for smart manufacturing.

Hands-On Activities

1. Design a digital twin of a smart factory or smart building.
2. Simulate predictive maintenance using sensor data.
3. Compare 5G and 6G use cases for smart cities.
4. Develop a connected monitoring dashboard.

Module V: Building Future Intelligent Digital Ecosystems

Developing smart healthcare systems by integrating XR, Digital Twins, AI and intelligent devices. Building autonomous smart manufacturing systems using Digital Twins, robotics and edge computing. Designing next-generation smart cities using 6G, satellite connectivity, IoT and immersive technologies. Developing sustainable engineering solutions using advanced manufacturing and intelligent computing. Exploring future applications including metaverse, Industry 5.0, autonomous systems, human-digital collaboration and intelligent digital ecosystems.

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Credits: 04
Contact Hours: 60

Course Title: Emerging Technologies
Maximum Marks : 100
Internal Evaluation : 40
External Evaluation : 60

Hands-On Activities

1. Design a smart city architecture integrating XR and 6G.
2. Develop a healthcare digital twin concept.
3. Create a smart manufacturing workflow integrating robotics and digital twins.
4. Mini Capstone Project: Build an application integrating two or more technologies

MINOR PROJECTS DURING THE SEMESTER

1. AI-Powered Personal Health and Wellness Assistant
2. Smart Energy Management using IoT and Predictive Analytics
3. Digital Twin-Based Smart Building Monitoring System
4. XR-Based Virtual Laboratory for Engineering Education
5. Blockchain-Enabled Smart Document Verification System

PEDAGOGY

The course follows a build and apply pedagogy that combines conceptual learning with practical implementation and real-world problem solving. Students will learn through interactive lectures, industry case studies, hands-on laboratory sessions, cloud-based platforms, simulations, guided programming exercises and collaborative mini-projects. The course emphasizes experiential learning using modern software tools, open-source technologies and emerging technology platforms. Continuous assessment will include quizzes, laboratory exercises, case studies, presentations, coding assignments and an application-oriented capstone project.

GUIDELINES FOR ASSESSMENT AND EVALUATION FOR SEMESTERS V - VIII OF FYUGP (DYD)

2. Assessment and Evaluation Model

Features of this Model of Assessment and Evaluation are:

- Domain knowledge: Tested through class tests and end-sem examination.
- Creativity: Assignments, seminars, open-ended examination questions.
- Hands-on experience: Practical tasks and project work to build real-world problem-solving skills.

Following scheme of assessment and evaluation shall be followed for the last two years of FYUGP (DYD):

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Semester VII
(For the session 2026, 2027, 2028)

Course Code: UMJDTT-713
Credits: 04
Contact Hours: 60

Course Title: Emerging Technologies
Maximum Marks : 100
Internal Evaluation : 40
External Evaluation : 60

1.1 Continuous Internal Assessment (CIA) - 40 Marks

This ensures sustained engagement and assessment beyond the final exam. CIA shall have the following components:

- Class Tests and Quizzes (10 Marks): This shall focus on concepts clarity, analytical ability comprising 2 or 3 short written or online tests.
- Assignments and Creative Tasks (10 Marks): This shall be focused on Problem-solving or reflective assignments and Open-ended questions to assess creativity.
- Practical and Hands-on Activities (15 Marks): This shall have the activities like Lab work, field tasks, coding assignments, design exercises, case study etc.
- Seminar and Presentation (5 Marks): This shall have individual or group presentation to be evaluated for communication, originality, creativity, and depth.

1.2 End-Semester Examination - 60 Marks

This is designed to test comprehensive domain knowledge and applications. The question paper shall have three sections:

- Section A - 20 Marks: Short answer questions to test fundamentals and breadth of domain knowledge.
- Section B - 20 Marks: Problem-solving and applications of the domain knowledge. The problems need to be conceptual numerical and case-based related to real world applications.
- Section C - 20 Marks: This is to be devoted to essay/analytical questions (choice-based) and have to be Open-ended, requiring synthesis, critical thinking, and creativity.

Recommended Textbooks

1. Bernard Marr – *Tech Trends in Practice: The 25 Technologies That Are Driving the 4th Industrial Revolution*, Wiley, 2020.
4. Klaus Schwab – *Shaping the Future of the Fourth Industrial Revolution*, Currency, 2018.
5. Bernard Marr – *Artificial Intelligence in Practice: How 50 Successful Companies Used AI and Machine Learning to Solve Problems*, Wiley, 2019.
6. David Chalmers, David Lewis and Mark R. Haag – *Cloud Computing: Implementation, Management, and Security*, CRC Press, 2020.

UNIVERSITY OF JAMMU
Four Year Innovative Undergraduate Program
(Design Your Degree)
B. Tech (Information Technology & Mathematical Innovations) (Revised)
Semester VII
(For the session 2026, 2027, 2028)

Course Code: UMJDTT-714	Course Title: Crypto Currency & Block Chain Technologies
Credits: 04	Maximum Marks : 100
Contact Hours: 60	Internal Evaluation : 40
	External Evaluation : 60

Course Description

This course introduces students to the rapidly evolving world of cryptocurrencies and blockchain technologies through a practical, application-oriented approach. Students will learn the fundamental concepts behind decentralized systems, cryptocurrencies, blockchain architecture, smart contracts, decentralized finance (DeFi), NFTs, and enterprise blockchain applications. Instead of focusing heavily on cryptography and mathematical proofs, the course emphasizes real-world use cases, hands-on blockchain development, wallet management, smart contract deployment, blockchain analytics, and decentralized application development. By the end of the course, students will be able to understand blockchain ecosystems, build simple blockchain-based applications, and evaluate the societal, financial, and industrial impact of decentralized technologies.

Learning Outcomes

1. Able to explain the principles of blockchain and cryptocurrency ecosystems.
2. Create and manage cryptocurrency wallets while understanding blockchain transactions.
3. Develop and deploy simple smart contracts and decentralized applications.
4. Analyze blockchain applications in finance, healthcare, supply chain, IoT, and digital identity.
5. Design a basic blockchain-based solution for solving real-world problems.

MODULE I: Foundations of Cryptocurrency and Blockchain Technologies

Understanding the evolution of digital money from financial systems to cryptocurrencies and decentralized finance. Introduction to blockchain technology, distributed ledgers, blocks, hashing, consensus mechanisms, and cryptocurrency ecosystems. Understanding how blockchain is transforming banking, supply chain, healthcare, education, government services, digital identity, and smart cities. Learning the fundamentals of decentralization, transparency, security, and trust in blockchain-enabled systems.

Hands-On Activities

1. Create and configure a cryptocurrency wallet using MetaMask.
2. Explore Bitcoin and Ethereum transactions using blockchain explorers.
3. Compare traditional databases with blockchain using a real-world use case.

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MODULE II: Cryptocurrency Ecosystem and Digital Asset Management

Understanding cryptocurrency transactions, wallet management, private and public keys, cryptocurrency exchanges, stablecoins, tokenization, NFTs, and digital asset ownership. Exploring Web3 applications, blockchain security, digital signatures, and common cryptocurrency frauds. Learning how blockchain enables secure peer-to-peer transactions, digital ownership, and decentralized financial services.

Hands-On Activities

1. Create and manage cryptocurrency wallets.
2. Perform testnet cryptocurrency transactions.
3. Explore NFT marketplaces and digital asset ownership.
4. Compare cryptocurrencies based on market capitalization and use cases.
5. Analyze blockchain transaction fees and network activity using Etherscan.

MODULE III: Smart Contracts and Decentralized Application Development

Learning the principles of smart contracts and decentralized applications (DApps). Developing smart contracts using Solidity and Remix IDE. Understanding Ethereum Virtual Machine (EVM), contract deployment, testing, and blockchain interaction using MetaMask. Exploring decentralized application architecture, token creation, and blockchain integration with web applications. Understanding security considerations and best practices in smart contract development.

Hands-On Activities

1. Develop and deploy a basic smart contract using Remix IDE.
2. Build a simple decentralized voting application.
3. Create a basic crowdfunding smart contract.
4. Connect MetaMask with Remix for smart contract deployment.
5. Test and verify deployed smart contracts on Ethereum Testnet.

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MODULE IV: Blockchain Applications Across Industries

Exploring blockchain solutions for banking, healthcare, supply chain management, agriculture, education, logistics, insurance and digital identity. Understanding blockchain-enabled traceability, secure medical records, academic certificate verification, land registration, smart cities, and transparent governance. Studying enterprise blockchain platforms and integration of blockchain with Artificial Intelligence and IoT for intelligent applications.

Hands-On Activities

1. Design a blockchain-based supply chain tracking system.
2. Develop a prototype for academic certificate verification.
3. Create a blockchain workflow for healthcare record management.
4. Analyze a real-world blockchain case study from industry.
5. Design a blockchain solution for a smart city application.

MODULE V: Emerging Trends in Blockchain and Real-World Innovation

Understanding blockchain entrepreneurship, startup ecosystems, ethical issues, cybersecurity challenges, regulatory frameworks, and future research directions. Learning how blockchain is enabling next-generation financial systems, digital governance, and decentralized digital economies with Layer-2 blockchain solutions, blockchain interoperability, and Web3 technologies.

Hands-On Activities

1. Explore a decentralized finance (DeFi) platform using a blockchain test network.
2. Analyze the governance model of a Decentralized Autonomous Organization (DAO).
3. Design a blockchain startup idea addressing a real-world problem.
4. Develop and present a blockchain-based application prototype.
5. Prepare a project report and demonstrate the developed blockchain solution.

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Minor Projects for this Semester

1. Decentralized Voting System : Build a secure blockchain-based voting platform ensuring transparency and voter privacy.
2. Medicine Supply Chain Tracking System : Design a blockchain solution to trace medicines from manufacturers to pharmacies, preventing counterfeit drugs.
3. Crowdfunding Platform using Smart Contracts : Create a decentralized crowdfunding platform with automated fund release based on predefined conditions.
4. Food Supply Chain Traceability System : Track agricultural or food products from farms to consumers using blockchain.
5. Blockchain-Based Digital Identity Management System : Create a decentralized identity verification system for secure authentication.
6. NFT-Based Digital Art Marketplace : Develop a basic marketplace for minting and trading digital artwork using NFTs.
7. Blockchain Wallet Transaction Analyzer : Build a dashboard to analyze blockchain wallet transactions, gas fees, and transaction history.
8. Carbon Credit Tracking using Blockchain : Design a transparent platform for recording and trading carbon credits.

Pedagogy

The course adopts an experiential, learner-centric, and project-based approach that integrates lectures with interactive discussions, case studies, demonstrations, hands-on laboratory sessions, blockchain simulation tools, smart contract development, collaborative learning, industry use cases, seminars, and application-oriented mini projects. Students will gain practical experience using blockchain platforms and development tools such as MetaMask, Remix IDE, Solidity, Ethereum Testnet, Ganache, and Etherscan**,** enabling them to design, develop, and evaluate blockchain-based solutions for real-world applications.

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GUIDELINES FOR ASSESSMENT AND EVALUATION FOR SEMESTERS V - VIII OF FYUGP (DYD)

1.Assessment and Evaluation Model

Features of this Model of Assessment and Evaluation are:

- Domain knowledge: Tested through class tests and end-sem examination.
- Creativity: Assignments, seminars, open-ended examination questions.
- Hands-on experience: Practical tasks and project work to build real-world problem-solving skills.

Following scheme of assessment and evaluation shall be followed for the last two years of FYUGP (DYD):

1.1 Continuous Internal Assessment (CIA) - 40 Marks

This ensures sustained engagement and assessment beyond the final exam. CIA shall have the following components:

- Class Tests and Quizzes (10 Marks): This shall focus on concepts clarity, analytical ability comprising 2 or 3 short written or online tests.
- Assignments and Creative Tasks (10 Marks): This shall be focused on Problem-solving or reflective assignments and Open-ended questions to assess creativity.
- Practical and Hands-on Activities (15 Marks): This shall have the activities like Lab work, field tasks, coding assignments, design exercises, case study etc.
- Seminar and Presentation (5 Marks): This shall have individual or group presentation to be evaluated for communication, originality, creativity, and depth.

1.2 End-Semester Examination - 60 Marks

This is designed to test comprehensive domain knowledge and applications. The question paper shall have three sections:

- Section A - 20 Marks: Short answer questions to test fundamentals and breadth of domain knowledge.

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Credits: 04	Maximum Marks : 100
Contact Hours: 60	Internal Evaluation : 40
	External Evaluation : 60

- Section B - 20 Marks: Problem-solving and applications of the domain knowledge. The problems need to be conceptual numerical and case-based related to real world applications.
- Section C - 20 Marks: This is to be devoted to essay/analytical questions (choice-based) and have to be Open-ended, requiring synthesis, critical thinking, and creativity.

Recommended Textbooks

1. Imran Bashir, *Mastering Blockchain*, 5th Edition, Packt Publishing, 2023.
2. Andreas M. Antonopoulos and Gavin Wood, *Mastering Ethereum: Building Smart Contracts and DApps*, O'Reilly Media, Latest Reprint Edition.
3. Narayan Prusty, *Building Blockchain Projects: Building Decentralized Blockchain Applications with Ethereum and Solidity*, Packt Publishing, Latest Edition.
4. Daniel Drescher, *Blockchain Basics: A Non-Technical Introduction in 25 Steps*, 2nd Edition, Apress, 2020.
5. Arshdeep Bahga and Vijay Madisetti, *Blockchain Applications: A Hands-on Approach*, VPT Publications, Latest Edition.

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Semester VII
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Course Code: UMJDTT-715
Credits: 04
Contact Hours: 60

Course Title: Robotics Process Automation & Drones
Maximum Marks : 100
Internal Evaluation : 40
External Evaluation : 60

Course Description

This course offers a comprehensive and practice-oriented introduction to Robotic Process Automation (RPA) and Drone Technologies, emphasizing real-world problem solving, automation, and intelligent system development. It covers the fundamentals of software automation using RPA tools, business process automation, workflow design, bots, AI-assisted automation, and drone technologies including UAV architecture, flight control systems, sensors, mission planning, autonomous navigation, and real-world drone applications.The course focuses on designing and implementing intelligent automation solutions for business and industrial processes while developing practical skills in drone operation, programming, aerial data collection, and autonomous missions. Students gain hands-on experience in automating repetitive digital tasks and integrating drones with sensors, computer vision, GPS, IoT, and cloud technologies for solving real-world challenges.

LEARNING OUTCOMES

After completing this course, students will be able to:

1. Understand the principles, architecture, and applications of Robotic Process Automation (RPA) and Drone Technologies.
2. Analyze real-world business and industrial problems and automate repetitive workflows using modern RPA tools.
3. Design, program, and operate drones by integrating sensors, GPS, flight controllers, and mission planning techniques.
4. Develop intelligent automation solutions by integrating RPA, AI, IoT, Computer Vision, and Drone technologies for practical applications.
5. Build, test, optimize, and deploy industry-oriented automation and drone-based projects that address real-world challenges.

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Credits: 04
Contact Hours: 60

Course Title: Robotics Process Automation & Drones
Maximum Marks : 100
Internal Evaluation : 40
External Evaluation : 60

MODULE I: Foundations of Intelligent Automation and Drone Systems
Fundamentals of Robotic Process Automation (RPA). Introduction to drones (UAVs): components, types, sensors, payloads, and applications. Understanding how Robotic Process Automation (RPA), robotics and drones are transforming banking, healthcare, manufacturing, agriculture, disaster management and smart cities. Collecting and monitoring operational data using IoT devices, drone sensors and robotic control systems. Understanding robotic systems using sensors, actuators, controllers, GPS, IMU, cameras and embedded systems.

Hands-On Activities

1. Identify automation opportunities in daily life, education, healthcare, banking, and business.
2. Explore a drone simulator and understand drone controls and flight modes.
3. Create a simple workflow diagram for automating a real-world business process.

MODULE II: Learning Robotic Process Automation and Drone Technologies

Exploring software robots, industrial robots, collaborative robots (Cobots) and unmanned aerial vehicles (UAVs) for real-world automation. Learning drone flight planning and autonomous navigation using flight controllers, GPS, IMU and waypoint navigation. Developing robot perception systems using cameras, ultrasonic sensors, LiDAR and obstacle detection. Understanding drone communication, telemetry and mission planning for intelligent robotic operations.

Hands-On Activities

1. Build an attendance automation workflow.
2. Automate invoice data extraction from PDF files.
3. Create an email automation workflow for report generation.
4. Develop a student result processing automation system.
5. Build an automated file management workflow.

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MODULE III: Building Intelligent Solutions using RPA and Robotics Platforms

Drone architecture. Flight control systems. Navigation using GPS and onboard sensors. Drone communication. Drone simulation, Image capture and aerial mapping. Building robotic perception systems using computer vision, image processing and sensor fusion. Developing robot monitoring and control systems using real-time communication and cloud integration. Integrating Artificial Intelligence for object detection, path planning, predictive maintenance and intelligent decision making.

Hands-On Activities

1. Plan and execute a drone mission using a simulator.
2. Capture aerial images and generate a simple map.
3. Detect objects in drone images using pre-trained AI models.
4. Monitor crop health or traffic using simulated drone imagery.

MODULE IV: AI-Powered Automation and Intelligent Drone Systems

Introduction to AI-enabled automation. Computer vision for drones. Object detection, image classification, and tracking. Intelligent document processing. Integrating AI services with RPA workflows. Integrating drones with cloud platforms. Designing smart inspection and infrastructure monitoring systems using computer vision, drones and defect detection techniques. Creating disaster management and emergency response systems using autonomous drones, thermal imaging and real-time communication.

Hands-On Activities:

1. Build an AI-powered invoice processing system using OCR and RPA.
2. Develop a drone image analysis application for crop or infrastructure monitoring.
3. Create an automation workflow triggered by AI-generated results.
4. Build a smart inspection system combining drone images and AI object detection.

MODULE V: Autonomous Robotics and Future Intelligent Systems

Developing autonomous robotic systems using localization, mapping, path planning and intelligent control. Building collaborative robotic and drone systems using multi-robot .

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Credits: 04	Maximum Marks : 100
Contact Hours: 60	Internal Evaluation : 40
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coordination, swarm intelligence and distributed decision making. Integrating Artificial Intelligence, IoT and cloud robotics for intelligent automation and remote operations. Exploring emerging applications including digital twins, Industry 5.0, autonomous delivery, humanoid robots and intelligent service robots.

Hands-On Activities:

1. Deploy a complete RPA workflow.
2. Demonstrate a drone-based monitoring application.
3. Evaluate the efficiency of an automation solution before and after deployment.
4. Present an integrated automation solution addressing a real-world problem.

MINOR PROJECTS DURING THE SEMESTER:

1. Autonomous Crop Monitoring and Reporting System
2. AI-Powered Infrastructure Inspection Platform
3. Smart Warehouse Inventory Monitoring using Drone Images
4. Drone-Based Search and Rescue Decision Support System
5. Intelligent Campus Surveillance System
6. Integrated Drone + RPA Inspection and Reporting Platform

PEDAGOGY:

The mentor shall introduce each Robotics Process Automation and Drone Technology concept by anchoring it to real-world applications such as banking automation, smart manufacturing, warehouse logistics, precision agriculture, infrastructure inspection, environmental monitoring, disaster management, and autonomous delivery systems. Concepts will be delivered through progressive laboratory sessions, simulations, industrial case studies, and project-based learning that closely resemble modern industrial workflows.

Students will continuously develop automation bots and drone-based solutions while learning problem analysis, workflow optimization, autonomous system design, sensor integration, AI-assisted decision making, and deployment strategies. Emphasis will be placed on system reliability, safety, regulatory compliance, teamwork, documentation, testing, and innovation. The teaching-learning process will adopt a "How to Solve It" problem-solving approach, enabling students to analyze complex real-world challenges, design intelligent automation solutions, and implement industry-ready systems.

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Course Title: Robotics Process Automation & Drones
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External Evaluation : 60

GUIDELINES FOR ASSESSMENT AND EVALUATION FOR SEMESTERS V - VIII OF FYUGP (DYD)

1.Assessment and Evaluation Model

Features of this Model of Assessment and Evaluation are:

- Domain knowledge: Tested through class tests and end-sem examination.
- Creativity: Assignments, seminars, open-ended examination questions.
- Hands-on experience: Practical tasks and project work to build real-world problem-solving skills.

Following scheme of assessment and evaluation shall be followed for the last two years of FYUGP (DYD):

1.1 Continuous Internal Assessment (CIA) - 40 Marks

This ensures sustained engagement and assessment beyond the final exam. CIA shall have the following components:

- Class Tests and Quizzes (10 Marks): This shall focus on concepts clarity, analytical ability comprising 2 or 3 short written or online tests.
- Assignments and Creative Tasks (10 Marks): This shall be focused on Problem-solving or reflective assignments and Open-ended questions to assess creativity.
- Practical and Hands-on Activities (15 Marks): This shall have the activities like Lab work, field tasks, coding assignments, design exercises, case study etc.
- Seminar and Presentation (5 Marks): This shall have individual or group presentation to be evaluated for communication, originality, creativity, and depth.

1.2 End-Semester Examination - 60 Marks

This is designed to test comprehensive domain knowledge and applications. The question paper shall have three sections:

- Section A - 20 Marks: Short answer questions to test fundamentals and breadth of domain knowledge.
- Section B - 20 Marks: Problem-solving and applications of the domain knowledge. The problems need to be conceptual numerical and case-based related to real world applications.

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- Section C - 20 Marks: This is to be devoted to essay/analytical questions (choice-based) and have to be Open-ended, requiring synthesis, critical thinking, and creativity.

RECOMMENDED TEXTBOOKS:

1. Alok Mani Tripathi – *Learning Robotic Process Automation*, Packt Publishing, 2018.
2. Tom Taulli – *The Robotic Process Automation Handbook*, Apress, 2020.
3. Richard Murdoch – *Robotic Process Automation: Guide to Building Software Robots*, Springer, 2018.
4. Frank Casale – *Introduction to Robotic Process Automation*, Institute of Robotic Process Automation, 2017.
5. Daniel Watzenig and Martin Horn – *Automated Driving and Autonomous Vehicles*, Springer, 2017.
6. Reg Austin – *Unmanned Aircraft Systems: UAV Design, Development and Deployment*, Wiley, 2010.
7. Paul G. Fahlstrom and Thomas Gleason – *Introduction to UAV Systems*, 4th Edition, Wiley, 2012.
8. Rafael C. Gonzalez and Richard E. Woods – *Digital Image Processing*, 4th Edition, Pearson, 2018.
9. Steven F. Barrett – *Arduino Microcontroller: Processing for Everyone!*, Morgan & Claypool, 2020.
10. Simon Monk – *Programming Arduino: Getting Started with Sketches*, 3rd Edition, McGraw-Hill, 2021.
11. Sebastian Thrun, Wolfram Burgard, and Dieter Fox – *Probabilistic Robotics*, MIT Press, 2005.
12. Klaus Schwab – *The Fourth Industrial Revolution*, Crown Business, 2017.

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Semester VII
(For the session 2026, 2027, 2028)

Course Code: UMJDTT-716 Course Title: AI in Neuro & Cognitive Computing
Credits: 04 Maximum Marks : 100
Contact Hours: 60 Internal Evaluation : 40
External Evaluation : 60

COURSE DESCRIPTION

This course introduces the fundamentals of Artificial Intelligence in Neuro and Cognitive Computing, focusing on how AI techniques can be applied to understand brain functions and cognitive processes. Students will learn the basics of neuroscience, brain signals, cognitive computing, EEG preprocessing, feature extraction, and AI techniques for cognitive computing applications. The course emphasizes hands-on learning through practical implementation and mini-projects.

LEARNING OUTCOMES

1. Understand the basic concepts of neuroscience, cognitive computing, and artificial intelligence.
2. Analyze EEG signals using preprocessing and feature extraction techniques.
3. Develop AI and deep learning models for cognitive computing applications.
4. Design simple neuro-inspired intelligent systems for real-world applications.

MODULE I: INTRODUCTION TO NEURO AND COGNITIVE COMPUTING

Artificial Intelligence (AI) and brain-inspired computing, biological and artificial neurons for intelligent model development, human brain structure and cognitive functions (memory, learning, attention, and decision making) for cognitive computing applications.

Hands-On Activities:

1. Explore brain anatomy using online visualization tools.
2. Build a simple Artificial Neural Network using Python.

MODULE II: BRAIN SIGNALS AND DATA ACQUISITION

Brain signal acquisition and Electroencephalography (EEG), EEG electrode placement using the 10–20 system, brain wave analysis (Delta, Theta, Alpha, Beta, and Gamma) for biomedical signal processing, public EEG datasets for AI-based research and model development, and their applications in healthcare and intelligent neurotechnology systems.

Hands-On Activities:

1. Visualize EEG signals using Python.
2. Explore public EEG datasets.

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Credits: 04
Contact Hours: 60

Course Title: AI in Neuro & Cognitive Computing
Maximum Marks : 100
Internal Evaluation : 40
External Evaluation : 60

MODULE III: EEG SIGNAL PREPROCESSING

EEG signal quality assessment and preprocessing for biomedical applications, noise and artifact analysis, signal filtering using band-pass and notch filters, artifact removal techniques for enhanced signal quality, signal segmentation (windowing) for machine learning and deep learning-based analysis, and normalization and standardization techniques for accurate and reliable predictive modeling.

Hands-On Activities:

1. Perform EEG preprocessing using Python.
2. Apply filtering and artifact removal techniques.
3. Prepare EEG data for analysis.

MODULE IV: FEATURE EXTRACTION FROM EEG SIGNALS

Feature extraction techniques for EEG signal analysis, time-domain feature extraction for physiological signal characterization, frequency-domain analysis using Fast Fourier Transform (FFT) and Power Spectral Density (PSD), time–frequency analysis using Short-Time Fourier Transform (STFT) and Wavelet Transform (WT), Differential Entropy (DE) for EEG feature representation, and feature selection techniques for enhancing the performance of machine learning and deep learning-based biomedical applications.

Hands-On Activities:

1. Extract time-domain and frequency-domain features from EEG signals.
2. Implement FFT and PSD using Python.
3. Compare different EEG feature extraction techniques.

MODULE V: AI FOR COGNITIVE COMPUTING AND ITS APPLICATIONS

Machine Learning in NeuroComputing, Artificial Neural Networks (ANN) models for intelligent prediction applications, Convolutional Neural Networks (CNN) models for computer vision and pattern recognition applications, Long Short-Term Memory (LSTM) models for sequential and time-series data analysis, deep learning applications in emotion recognition, cognitive state assessment, and Brain–Computer Interface (BCI) systems, along with model performance evaluation and comparative analysis.

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Credits: 04
Contact Hours: 60

Course Title: AI in Neuro & Cognitive Computing
Maximum Marks : 100
Internal Evaluation : 40
External Evaluation : 60

Hands-On Activities:

1. Develop a basic CNN or LSTM model for EEG classification.
2. Evaluate the performance of AI models using standard metrics.

MINOR PROJECTS DURING THE SEMESTER

1. Develop a simple EEG signal visualization system using Python.
2. Design an EEG signal preprocessing and noise removal tool.
3. Implement EEG feature extraction using FFT and Power Spectral Density (PSD).
4. Develop a basic emotion recognition system using machine learning.
5. Build a simple Brain–Computer Interface (BCI) application using public EEG dataset.

PEDAGOGY

Mentor should introduce each topic through real-world examples from healthcare, brain–computer interfaces, and intelligent systems to help students understand the practical relevance of Neuro and Cognitive Computing. The course should emphasize experiential learning through hands-on activities, Python-based programming, EEG signal analysis, and simple AI model development. Students should be encouraged to learn by exploring public datasets, implementing algorithms, and solving application-oriented problems using the "How to Solve It" approach. The focus should be on developing analytical thinking, problem-solving skills, and practical experience rather than memorizing theoretical concepts.

GUIDELINES FOR ASSESSMENT AND EVALUATION FOR SEMESTERS V - VIII OF FYUGP (DYD)

1.Assessment and Evaluation Model

Features of this Model of Assessment and Evaluation are:

- Domain knowledge: Tested through class tests and end-sem examination.
- Creativity: Assignments, seminars, open-ended examination questions.
- Hands-on experience: Practical tasks and project work to build real-world problem-solving skills.

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Course Code: UMJDTT-716
Credits: 04
Contact Hours: 60

Course Title: AI in Neuro & Cognitive Computing
Maximum Marks : 100
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Following scheme of assessment and evaluation shall be followed for the last two years of FYUGP (DYD):

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- Assignments and Creative Tasks (10 Marks): This shall be focused on Problem-solving or reflective assignments and Open-ended questions to assess creativity.
- Practical and Hands-on Activities (15 Marks): This shall have the activities like Lab work, field tasks, coding assignments, design exercises, case study etc.
- Seminar and Presentation (5 Marks): This shall have individual or group presentation to be evaluated for communication, originality, creativity, and depth.

1.2 End-Semester Examination - 60 Marks

This is designed to test comprehensive domain knowledge and applications. The question paper shall have three sections:

- Section A - 20 Marks: Short answer questions to test fundamentals and breadth of domain knowledge.
- Section B - 20 Marks: Problem-solving and applications of the domain knowledge. The problems need to be conceptual numerical and case-based related to real world applications.
- Section C - 20 Marks: This is to be devoted to essay/analytical questions (choice-based) and have to be Open-ended, requiring synthesis, critical thinking, and creativity.

RECOMMENDED TEXTBOOKS

1. Satish Kumar, Neurocomputing: Principles and Applications, Springer.
2. Rajesh P. N. Rao, Brain-Computer Interfacing: An Introduction, Cambridge University Press.
3. Michael A. Arbib (Ed.), The Handbook of Brain Theory and Neural Networks, MIT Press.
4. Stephen Jose Hanson & Read Montague, Computational Neuroscience, MIT Press.
5. Michael W. Eysenck & Mark T. Keane, Cognitive Psychology: A Student's Handbook, Routledge.
6. Stuart Russell & Peter Norvig, Artificial Intelligence: A Modern Approach, Pearson.

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Semester VII
(For the session 2026, 2027, 2028)

Course Code: UMJDTT-717	Course Title: Numerical methods: from approximation to perfection
Credits: 04	Maximum Marks : 100
Contact Hours: 60	Internal Evaluation : 40
	External Evaluation : 60

COURSE DESCRIPTION

This course introduces students to systematic techniques for obtaining approximate solutions to mathematical problems that are difficult or impossible to solve analytically. In modern computing, artificial intelligence, data science, simulation, and engineering analysis, exact answers are often unavailable, unnecessary, or computationally expensive. This course develops the ability to formulate, compute, analyze, and interpret numerical solutions for equations, functions, data models, linear systems, and differential equations. With a balanced emphasis on theory, algorithmic thinking, and implementation, the course prepares students to apply numerical computation in AI systems, scientific modelling, intelligent decision support, robotics, graphics, and other technology-driven domains.

LEARNING OUTCOMES

The objectives of the course on "Numerical Methods: From Approximation to Perfection" include:

- understanding the role of numerical approximation in scientific and computational problem-solving, and identifying the major sources of numerical error, including truncation, round-off, and floating-point limitations;
- solving nonlinear equations using classical numerical root-finding methods, analyzing convergence behavior, and selecting suitable stopping criteria;
- applying interpolation, spline ideas, and least squares curve fitting to estimate values, represent data trends, and model information from discrete or noisy datasets;
- performing numerical differentiation and numerical integration, and evaluating the accuracy and error characteristics of the computed results;
- solving systems of linear equations using direct and iterative methods such as Gauss elimination, LU decomposition, Jacobi, and Gauss-Seidel techniques;
- introducing eigenvalue- and eigenvector-based numerical ideas for matrix computation, data analysis, and computational modelling;
- applying numerical methods for ordinary differential equations to practical problems in AI, IT, simulation, robotics, and scientific computing, using appropriate computational tools.

PREREQUISITES: Students are expected to possess a basic understanding of calculus, algebra, matrices, and elementary programming concepts.

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Course Title: Numerical methods: from approximation to perfection
Credits: 04
Maximum Marks : 100
Contact Hours: 60
Internal Evaluation : 40
External Evaluation : 60

MODULE I: APPROXIMATION AND ERRORS – WHY EXACT ANSWERS ARE NOT ALWAYS POSSIBLE

In computing, every number stored in memory has a limit, and every calculation carries some degree of uncertainty. A weather forecast, a machine-learning model, or a robotics control system rarely depends on an exact symbolic answer; instead, it depends on a dependable approximation that is computed efficiently and interpreted correctly. This module introduces the foundations of numerical computation by explaining why approximation is essential and how error influences the quality of a numerical result.

KEY CONCEPTS:

- Need and scope of numerical methods in computing, science, and engineering
- Approximation as a practical strategy in numerical problem-solving
- Types of errors: absolute error, relative error, percentage error
- Truncation error and round-off error
- Significant digits and decimal precision
- Floating-point representation and limitations of computer arithmetic
- Sources of computational error in iterative and algebraic calculations
- Computational cost, memory constraints, and precision trade-offs

Hands-On Activities

1. Compare exact and floating-point representations of decimal numbers using Python and demonstrate round-off effects.
2. Use a Jupyter Notebook to evaluate truncation error in successive approximations of a mathematical expression or function.

MODULE II: FINDING ROOTS – NUMERICAL SOLVERS FOR EQUATIONS

A sensor model, a network balance equation, or an optimization condition often leads to a nonlinear equation whose solution cannot be written in a convenient closed form. Numerical root-finding methods provide reliable iterative procedures for locating such solutions. This module trains students to understand and implement classical root-solving algorithms and to select an appropriate method based on speed, stability, and convergence.

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KEY CONCEPTS:

- Formulation of root-finding problems in engineering and computational models
- Bisection method: principle, algorithm, and convergence
- False position method and comparison with bisection
- Fixed-point iteration: transformation and iteration mapping
- Newton-Raphson method: derivation, implementation, and efficiency
- Convergence, divergence, and rate of convergence
- Initial guess selection and stopping criteria
- Applications in AI calibration, control models, and engineering equations

Hands-On Activities

1. Implement bisection, false position, and Newton-Raphson methods in Python for selected nonlinear equations.
2. Use GeoGebra or Desmos to visualize iterative convergence and compare the effect of different initial guesses.

MODULE III: INTERPOLATION, CURVE FITTING, AND APPROXIMATION FROM DATA

In data-driven environments, values are frequently known only at specific points from experiments, sensors, logs, or measurements. A system may need intermediate estimates, trend analysis, or smooth approximations before the data can be used effectively. This module focuses on interpolation and curve fitting methods that transform discrete observations into useful computational models.

KEY CONCEPTS:

- Concept of interpolation and approximation from observed data
- Difference tables and polynomial interpolation framework
- Newton’s forward interpolation formula
- Newton’s backward interpolation formula
- Lagrange interpolation formula and applications
- Spline basics and piecewise smooth approximation
- Least squares curve fitting for linear and nonlinear trends
- Approximation using noisy data and model selection considerations

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Hands-On Activities

1. Use Python/NumPy to interpolate temperature, sales, or sensor data using Newton and Lagrange methods.
2. Fit a least squares curve to noisy data in MATLAB or Jupyter Notebook and compare the fit with the original observations.

MODULE IV: NUMERICAL DIFFERENTIATION, INTEGRATION, AND LINEAR SYSTEMS

Many scientific and AI computations require the evaluation of rates of change, accumulated quantities, and large systems of equations. Since exact integration or symbolic manipulation is not always possible, numerical schemes offer workable and efficient alternatives. This module develops techniques for differentiation, integration, and the solution of linear systems, which form the backbone of computational problem-solving.

KEY CONCEPTS:

- Numerical differentiation from tabulated or sampled data
- Forward, backward, and central difference approximations
- Trapezoidal rule for numerical integration
- Simpson's 1/3 rule and Simpson's 3/8 rule
- Error analysis in numerical integration
- System of linear equations in computational applications
- Direct methods: Gauss elimination and pivoting
- Matrix factorization and LU decomposition
- Iterative methods: Jacobi and Gauss-Seidel methods
- Convergence criteria and computational efficiency

Hands-On Activities

1. Compute numerical derivatives and definite integrals for sample functions using Python/SciPy and compare the results with exact values where available.
2. Solve a system of linear equations using Gauss elimination, LU decomposition, Jacobi, and Gauss-Seidel methods in a MATLAB or Python notebook.

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	External Evaluation : 60

MODULE V: DIFFERENTIAL EQUATIONS, EIGEN PROBLEMS, AND COMPUTATIONAL APPLICATIONS

Dynamic systems in robotics, signal processing, simulation, and scientific computing are frequently modeled through differential equations and matrix-based computational structures. This module introduces numerical techniques for ordinary differential equations and builds a conceptual bridge to eigenvalue computations, allowing students to appreciate how numerical methods support intelligent systems and large-scale simulations.

KEY CONCEPTS:

- Numerical solution of ordinary differential equations (ODEs)
- Initial value problems and stepwise approximation
- Euler method and its limitations
- Improved Euler method and Runge-Kutta methods
- Stability, accuracy, and step-size control
- Introduction to eigenvalues and eigenvectors in numerical computation
- Numerical relevance of eigen problems in data analysis and system modeling
- Applications in AI, simulation, robotics, graphics, and scientific computing
- Case-based implementation for real-world computational models

Hands-On Activities

1. Simulate an ODE using Euler and Runge-Kutta methods in Python and compare stability and accuracy.
2. Use NumPy/SciPy to compute eigenvalues and eigenvectors for a matrix model arising from graph, vibration, or data-analysis contexts.

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MINOR PROJECTS DURING THE SEMESTER

1. Traffic Flow Approximation Model – Develop a numerical model for estimating traffic density or travel time using time-series data from a local road corridor.
2. Weather or Climate Trend Estimator – Use interpolation and curve fitting to approximate temperature, rainfall, or humidity trends from regional weather data.
3. Signal Smoothing and Noise Reduction Tool – Build a small application for smoothing noisy sensor, audio, or biomedical signals using numerical approximation and least squares fitting.
4. Robot Path Simulation – Simulate a mobile robot trajectory using numerical methods for differential equations and evaluate path stability under changing step sizes.
5. Smart Campus or Healthcare Prediction Model – Create a numerical prototype for estimating energy usage, attendance trends, patient queue variation, or resource demand using data-driven approximation.

GUIDELINES FOR ASSESSMENT AND EVALUATION FOR SEMESTERS V - VIII OF FYUGP (DYD)

1.Assessment and Evaluation Model

Features of this Model of Assessment and Evaluation are:

- Domain knowledge: Tested through class tests and end-sem examination.
- Creativity: Assignments, seminars, open-ended examination questions.
- Hands-on experience: Practical tasks and project work to build real-world problem-solving skills.

Following scheme of assessment and evaluation shall be followed for the last two years of FYUGP (DYD):

1.1 Continuous Internal Assessment (CIA) - 40 Marks

This ensures sustained engagement and assessment beyond the final exam. CIA shall have the following components:

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- Class Tests and Quizzes (10 Marks): This shall focus on concepts clarity, analytical ability comprising 2 or 3 short written or online tests.
- Assignments and Creative Tasks (10 Marks): This shall be focused on Problem-solving or reflective assignments and Open-ended questions to assess creativity.
- Practical and Hands-on Activities (15 Marks): This shall have the activities like Lab work, field tasks, coding assignments, design exercises, case study etc.
- Seminar and Presentation (5 Marks): This shall have individual or group presentation to be evaluated for communication, originality, creativity, and depth.

1.2 End-Semester Examination - 60 Marks

This is designed to test comprehensive domain knowledge and applications. The question paper shall have three sections:

- Section A - 20 Marks: Short answer questions to test fundamentals and breadth of domain knowledge.
- Section B - 20 Marks: Problem-solving and applications of the domain knowledge. The problems need to be conceptual numerical and case-based related to real world applications.
- Section C - 20 Marks: This is to be devoted to essay/analytical questions (choice-based) and have to be Open-ended, requiring synthesis, critical thinking, and creativity.

RECOMMENDED TEXTBOOKS

1. S. S. Sastry, *Introductory Methods of Numerical Analysis*, PHI Learning.
2. M. K. Jain, S. R. K. Iyengar, and R. K. Jain, *Numerical Methods for Scientific and Engineering Computation*, New Age International.
3. Chapra, Steven C. and Canale, Raymond P., *Numerical Methods for Engineers*, McGraw-Hill.
4. Burden, Richard L. and Faires, J. Douglas, *Numerical Analysis*, Cengage Learning.

Optional Digital / Video / Project Resources

1. Python scientific stack: NumPy, SciPy, Matplotlib, Jupyter Notebook
2. MATLAB documentation and tutorials
3. GeoGebra and Desmos for function visualization and iterative method exploration
4. NPTEL / SWAYAM lectures on numerical methods and scientific computing
5. Open datasets from meteorology, transportation, and smart-city platforms for mini-project work.

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Course Code: UMJDTP-718	Course Title : Project 1
Credits: 06	Maximum Marks : 150
	Internal Evaluation : 100
	External Evaluation : 50

Guidelines for Interdisciplinary Major Projects

To promote transdisciplinarity, experiential learning, skill development and local to global relevance the essence of the DYD program the BOS of SIIEDC resolves to have at least two semester long major projects carrying 150 marks each in semesters V to VIII.

1. Objectives

a) Integrate theoretical knowledge across disciplines.
b) Encourage research-oriented and inquiry-driven learning.
c) Develop analytical, digital, and communication competencies.
d) local, regional, or global developmental challenges.
e) Enhance employability and innovation skills.

2. Nature and Scope of Projects

Projects may include interdisciplinary academic research, mathematical or computational modelling, design thinking and technological solutions, community engagement initiatives, policy analysis of educational innovation and entrepreneurship or sustainability-focused projects. Each project must clearly demonstrate integration of at least two courses of the current semester. Transdisciplinary project in collaboration are also encouraged.

3. Structure and Supervision

A group of 3-5 students shall work on the project where interdisciplinary team formation is encouraged. One Primary Mentors shall guide the project at least one from another discipline(s)/course(s) may be appointed where required. The project shall be conducted over one semester with the following suggested indicators:

a) Proposal Development and Approval
b) Literature Review and Conceptual Framework
c) Methodology Design
d) Implementation/Analysis
e) Mid-Term Review
f) Final Report Submission
g) Presentation and Viva-Voce

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Course Code: UMJDTP-718
Credits: 06

Course Title : Project 1
Maximum Marks : 150
Internal Evaluation : 100
External Evaluation : 50

4. Proposal Guidelines

Each project proposal shall include

a) Title and thematic area.
b) Courses/disciplines integrated.
c) Problem statement and motivation.
d) Objectives and expected outcomes.
e) Brief literature review.
f) Methodology and tools.
g) Timeline with milestones,
h) Ethical considerations (if applicable).

Approval shall be granted by the DYD Academic/Project Committee.

5. Evaluation Scheme (Credits-6) Marks: 150

a) Continuous Assessment - 40 Marks

Proposal quality and originality	10
Literature review	10
Methodology planning	10
Progress, logbook & mentor assessment	10

b) Final Report - 60 Marks

Problem formulation and clarity	10
Interdisciplinary integration	10
Methodology and analysis	20
Innovation and societal/technological relevance	10
Writing quality and references	10

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Course Code: UMJDTP-718
Credits: 06

Course Title : Project 1
Maximum Marks : 150
Internal Evaluation : 100
External Evaluation : 50

c) Presentation & Viva Voce-50 Marks

Conceptual understanding	15
Implementation/results/outcome	15
Communication skills	5
Response to questions	10
Collaboration/team work	5

Students shall submit a hardbound copy of the project report, along with a soft copy containing the PDF of the report and all relevant datasets, source code, and design artefacts. In addition, students mast submit the presentation slides prepared for the project evaluation and a duly maintained project diary/logbook certified by the mentor(s).

6. Academic Integrity and Ethics

To meet academic integrity and professional ethics, proper citation and avoidance of plagiarism are mandatory. Also responsible use of Al tools must be declared.

7. Role of the DYD Project Committee

The committee shall approve project proposals, ensure interdisciplinary balance, monitor the progress. standardize evaluation procedures and maintain a repository of DYD projects.

8. Timeline of the Project:

The project work is required to be initiated in the following phases:
Week 1-3: Proposal preparation and approval
Week 4-6: Literature review, data collection and methodology
Week 8-12: Implementation/Analysis
Week 13: Final submission and viva-voce.

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Course Code: UMJDTP-718	Course Title : Project 1
Credits: 06	Maximum Marks : 150
	Internal Evaluation : 100
	External Evaluation : 50

9. Outcomes:

By the end of 8th semester, the students shall have atleast any one of the following outcomes:

a) Application for Intellectual Property Rights (IPR) such as design patent, copyright, etc.
b) Submission of a high-quality research article in a Scopus/SCI/SCIE indexed journal, with the manuscript accepted for review/publication.
c) Conversion of the idea into a start-up, including submission of an application to an incubator for start-up support.

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Course Code: UMJDTP-719	Course Title : Project 2
Credits: 06	Maximum Marks : 150
	Internal Evaluation : 100
	External Evaluation : 50

Guidelines for Interdisciplinary Major Projects

To promote transdisciplinarity, experiential learning, skill development and local to global relevance the essence of the DYD program the BOS of SIIEDC resolves to have at least two semester long major projects carrying 150 marks each in semesters V to VIII.

1. Objectives

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c) Develop analytical, digital, and communication competencies.
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e) Enhance employability and innovation skills.

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Projects may include interdisciplinary academic research, mathematical or computational modelling, design thinking and technological solutions, community engagement initiatives, policy analysis of educational innovation and entrepreneurship or sustainability-focused projects. Each project must clearly demonstrate integration of at least two courses of the current semester. Transdisciplinary project in collaboration are also encouraged.

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A group of 3-5 students shall work on the project where interdisciplinary team formation is encouraged. One Primary Mentors shall guide the project at least one from another discipline(s)/course(s) may be appointed where required. The project shall be conducted over one semester with the following suggested indicators:

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Course Code: UMJDTP-719
Credits: 06

Course Title : Project 2
Maximum Marks : 150
Internal Evaluation : 100
External Evaluation : 50

4. Proposal Guidelines

Each project proposal shall include

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b) Courses/disciplines integrated.
c) Problem statement and motivation.
d) Objectives and expected outcomes.
e) Brief literature review.
f) Methodology and tools.
g) Timeline with milestones,
h) Ethical considerations (if applicable).

Approval shall be granted by the DYD Academic/Project Committee.

5. Evaluation Scheme (Credits-6) Marks: 150

a) Continuous Assessment - 40 Marks

Proposal quality and originality	10
Literature review	10
Methodology planning	10
Progress, logbook & mentor assessment	10

b) Final Report - 60 Marks

Problem formulation and clarity	10
Interdisciplinary integration	10
Methodology and analysis	20
Innovation and societal/technological relevance	10
Writing quality and references	10

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Credits: 06

Course Title : Project 2
Maximum Marks : 150
Internal Evaluation : 100
External Evaluation : 50

c) Presentation & Viva Voce-50 Marks

Conceptual understanding	15
Implementation/results/outcome	15
Communication skills	5
Response to questions	10
Collaboration/team work	5

Students shall submit a hardbound copy of the project report, along with a soft copy containing the PDF of the report and all relevant datasets, source code, and design artefacts. In addition, students mast submit the presentation slides prepared for the project evaluation and a duly maintained project diary/logbook certified by the mentor(s).

6. Academic Integrity and Ethics

To meet academic integrity and professional ethics, proper citation and avoidance of plagiarism are mandatory. Also responsible use of AI tools must be declared.

7. Role of the DYD Project Committee

The committee shall approve project proposals, ensure interdisciplinary balance, monitor the progress. standardize evaluation procedures and maintain a repository of DYD projects.

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The project work is required to be initiated in the following phases:
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By the end of 8th semester, the students shall have atleast any one of the following outcomes:

a) Application for Intellectual Property Rights (IPR) such as design patent, copyright, etc.
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Course Code: UMJDTT-811
Credits: 04
Contact Hours: 60

Course Title: Advance, Generative AI & Agents
Maximum Marks : 100
Internal Evaluation : 40
External Evaluation : 60

COURSE DESCRIPTION

This course provides a practical introduction to Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Multimodal AI, and Agentic AI through hands-on development of real-world AI applications. The course follows a "Build While Learning" approach where students progressively develop AI-powered systems using modern AI platforms, APIs, open-source frameworks, and workflow automation tools. Students will learn prompt engineering, build intelligent chatbots, develop document-aware AI assistants using RAG and vector databases, and design autonomous AI agents capable of reasoning, planning, tool usage, and workflow automation using different frameworks. By the end of the course, students will have built multiple deployable AI applications and an industry-oriented capstone project demonstrating practical problem-solving using modern Generative AI technologies.

LEARNING OUTCOMES

By the end of this course, students will be able to:

1. Build practical Generative AI applications using modern LLM APIs and open-source models.
2. Design effective prompts for solving real-world tasks using advanced prompting techniques.
3. Develop Retrieval-Augmented Generation applications integrated with vector databases and custom knowledge bases.
4. Build autonomous AI agents using CrewAI, BeeAI, AutoGen, and workflow automation tools.
5. Deploy, evaluate, and improve AI applications while applying responsible AI practices.

MODULE I: AI Foundations through Real-World Data and Intelligent Problem Solving

Understanding how AI is transforming healthcare, finance, agriculture, cybersecurity, education and scientific research. AI vs Data Science vs Machine Learning vs Deep Learning vs Generative AI. Understanding different types of data through real-world datasets (text, images, audio, sensor, medical, satellite and business data). AI for Multimodal Applications: Develop applications that understand and generate text, images, audio and visual content for education, healthcare, accessibility and media.

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Hands-On Activities

1. Compare the responses of ChatGPT, Gemini, Claude, and Perplexity for the same prompt and analyze differences.
2. Build a personal AI study assistant for summarizing notes, generating quizzes, and creating flashcardsDevelop a simple AI-powered FAQ chatbot using a no-code platform.
3. Design prompts using different prompting techniques and compare the generated outputs.

MODULE II: Generative AI, AI Assistants and Intelligent Automation.

AI-assisted Software Development: Using GitHub Copilot, Cursor AI, Windsurf for code generation, debugging, documentation, code review and software testing. AI for Intelligent Search and Knowledge Discovery: Using Perplexity AI, NotebookLM and ChatGPT Search for fact-based search, knowledge synthesis, document understanding and question answering.
AI for Workflow Automation and Productivity: Using Zapier AI, and crewAI for intelligent workflow automation, email processing, document management and business process automation. AI Prompt Engineering : Designing effective prompts for software development, education, healthcare, finance, customer support, research and business decision-making.

Hands-On Activities

1. Develop an AI-powered chatbot using a no-code AI platform.
2. Build a document Question-Answering application by connecting AI to PDF documents or a custom knowledge base.
3. Design an automated workflow using AI to summarize emails, generate meeting notes, or process documents.
4. Build an AI interview preparation assistant with personalized questions.

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MODULE III : Building Intelligent AI Applications using Large Language Models (LLMs)

Building AI Chatbots and Virtual Assistants: Develop conversational AI systems for customer support, education, healthcare and business applications using ChatGPT API, Gemini API, Hugging Face models and open-source LLMs.Developing Document-Aware AI Systems: Build intelligent assistants capable of understanding PDFs, reports, policies, legal documents and research papers using Retrieval-Augmented Generation (RAG), vector databases and semantic search. Building AI Applications using Modern Frameworks: Develop LLM-powered applications using LangChain, LangGraph, LlamaIndex.

Hands-On Activities

1. Build a PDF Question-Answering application using RAG.
2. Develop a university information assistant using syllabus, timetable, and notices.
3. Create a research paper assistant capable of answering questions from academic papers.
4. Build a company policy or employee handbook assistant using a custom knowledge base.

MODULE IV : Designing Autonomous AI Agents and Multi-Agent Systems for Real-World Automation

AI Agents for Real-World Applications: Build intelligent agents capable of reasoning, planning, memory, tool usage and decision making for solving real-world problems. Multi-Agent Systems and AI Orchestration: Design collaborative AI systems where multiple agents coordinate and perform complex tasks using CrewAI, BeeAI, AutoGen and LangGraph. AI Agents with External Tools: Develop agents that interact with APIs, databases, search engines, spreadsheets and cloud services to perform real-world tasks.

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Hands-On Activities

1. Automate a Business Workflow using Zapier, Make, or n8n by integrating AI with Gmail, Google Drive, Slack, Microsoft Teams, or spreadsheets.
2. Build a Multi-Agent Content Creation System where multiple AI agents collaboratively perform research, writing, editing, and proofreading for blogs or reports.
3. Create an AI Email Automation Agent that classifies emails, drafts replies, prioritizes messages, and organizes tasks.

MODULE V: Deploying Responsible AI Solutions and Industry Capstone Applications

Deploying AI Applications: Deploy AI solutions using Streamlit, Gradio, Hugging Face Spaces, Docker and cloud-based AI platforms. Monitoring and Improving AI Systems: Evaluate AI applications using user feedback, performance metrics, safety testing and continuous improvement techniques. Responsible and Ethical AI: Apply fairness, transparency, explainability, privacy, copyright, AI governance and responsible AI principles while developing intelligent systems. Emerging Trends in Generative AI: Explore AI copilots, AI operating systems, on-device AI, edge AI and multimodal agents.

Hands-On Activities

1. Evaluate AI-generated outputs for accuracy, bias, and reliability.
2. Deploy an AI application using Streamlit or Gradio.
3. Prepare and present a deployable AI application with documentation and ethical considerations.

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MINOR PROJECTS DURING THE SEMESTER

1. AI Research Assistant capable of answering questions from research papers using Retrieval-Augmented Generation (RAG).
2. Smart Campus AI Assistant for answering university-related queries, timetable information, notices, and FAQs.
3. AI Resume Analyzer and Career Advisor that reviews resumes, suggests improvements, and recommends career paths.
4. Multi-Agent Personal Productivity System that automates scheduling, email drafting, report generation, meeting summarization, and task management.
5. Multimodal AI Tutor capable of explaining concepts using text, images, diagrams, and interactive question answering.
6. Domain-Specific AI Chatbot (Healthcare, Banking, Agriculture, Tourism, Legal Assistance, or Education) powered by custom knowledge bases and RAG.

PEDAGOGY:

The course will be delivered through an experiential, hands-on learning approach that emphasizes learning by building rather than memorizing theoretical concepts. Each topic should be introduced using practical, real-world applications such as AI-powered chatbots, content generation, coding assistants, document analysis, smart search systems, and autonomous AI agents. Students should explore Generative AI through guided experimentation with modern AI tools and development platforms, following a "Learn, Build, Evaluate, Improve" methodology. Classroom sessions should integrate live demonstrations, prompt design exercises, mini-projects, collaborative activities, case studies, and application development, enabling students to design, evaluate, and deploy AI-powered solutions for real-world problems while understanding responsible and ethical AI practices.

UNIVERSITY OF JAMMU
Four Year Innovative Undergraduate Program
(Design Your Degree)
B. Tech (Information Technology & Mathematical Innovations) (Revised)
Semester VIII
(For the session 2026, 2027, 2028)

Course Code: UMJDTT-811
Credits: 04
Contact Hours: 60

Course Title: Advance, Generative AI & Agents
Maximum Marks : 100
Internal Evaluation : 40
External Evaluation : 60

GUIDELINES FOR ASSESSMENT AND EVALUATION FOR SEMESTERS V - VIII OF FYUGP (DYD)

1.Assessment and Evaluation Model

Features of this Model of Assessment and Evaluation are:

- Domain knowledge: Tested through class tests and end-sem examination.
- Creativity: Assignments, seminars, open-ended examination questions.
- Hands-on experience: Practical tasks and project work to build real-world problem-solving skills.

Following scheme of assessment and evaluation shall be followed for the last two years of FYUGP (DYD):

1.1 Continuous Internal Assessment (CIA) - 40 Marks

This ensures sustained engagement and assessment beyond the final exam. CIA shall have the following components:

- Class Tests and Quizzes (10 Marks): This shall focus on concepts clarity, analytical ability comprising 2 or 3 short written or online tests.
- Assignments and Creative Tasks (10 Marks): This shall be focused on Problem-solving or reflective assignments and Open-ended questions to assess creativity.
- Practical and Hands-on Activities (15 Marks): This shall have the activities like Lab work, field tasks, coding assignments, design exercises, case study etc.
- Seminar and Presentation (5 Marks): This shall have individual or group presentation to be evaluated for communication, originality, creativity, and depth.

1.2 End-Semester Examination - 60 Marks

This is designed to test comprehensive domain knowledge and applications. The question paper shall have three sections:

- Section A - 20 Marks: Short answer questions to test fundamentals and breadth of domain knowledge.
- Section B - 20 Marks: Problem-solving and applications of the domain knowledge. The problems need to be conceptual numerical and case-based related to real world applications.
- Section C - 20 Marks: This is to be devoted to essay/analytical questions (choice-based) and have to be Open-ended, requiring synthesis, critical thinking, and creativity.

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RECOMMENDED TEXTBOOKS

1. Nikhil Abraham, *Building LLM Powered Applications*, Packt Publishing, 2024.
2. Noah Gift & Alfredo Deza, *Generative AI on AWS*, O'Reilly Media, 2024. *(Excellent for practical AI application development.)*
3. Ben Auffarth, *Generative AI with Python and PyTorch*, Packt Publishing, 2024.
4. Chris Fregly & Antje Barth, *Generative AI on AWS: Building Context-Aware Multimodal Reasoning Applications*, O'Reilly Media, 2024.
5. Jay Alammar & Maarten Grootendorst, *Hands-On Large Language Models*, O'Reilly Media, 2024. *(Highly recommended for undergraduate students.)*
6. Sebastian Raschka, *Build a Large Language Model (From Scratch)*, Manning Publications, 2024. *(Reference only—not all chapters are required.)*
7. Ian Goodfellow, Yoshua Bengio & Aaron Courville, *Deep Learning*, MIT Press, 2016. *(Reference for AI fundamentals.)*

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Course Code: UMJDTT-812
Credits: 04
Contact Hours: 60

Course Title: The Science of Chance
Maximum Marks : 100
Internal Evaluation : 40
External Evaluation : 60

COURSE DESCRIPTION

Modern computing systems operate under uncertainty. Search engines rank uncertain information, autonomous vehicles make probabilistic decisions, cybersecurity systems detect anomalous behavior, healthcare AI estimates disease risk, and financial systems predict future trends. This course introduces the science of chance through probability, statistics, simulation, and data-driven decision making using real-world IT applications. This course focuses on understanding uncertainty through practical experimentation, programming, simulations, and case studies. Students will learn how randomness influences modern technologies and how statistical thinking forms the foundation of Artificial Intelligence, Machine Learning, Data Science, Networks, Cybersecurity, and Software Engineering. The course integrates Python-based hands-on activities, visualization, simulations, and real datasets to build intuition before introducing mathematical formulations.

LEARNING OUTCOMES

After successful completion of the course, students will be able to:

1. Explain randomness, uncertainty, and probability in engineering systems.
2. Apply probability and statistical methods to analyze real-world datasets.
3. Perform exploratory data analysis using visualization and statistical measures.
4. Design simulation-based solutions for uncertain systems using Python.
5. Interpret statistical evidence for decision making in AI, networking, cybersecurity, healthcare, and business applications.
6. Develop small analytical projects using probability, statistics, and simulation techniques.

Module I : Understanding Uncertainty in the Digital World

Randomness and Deterministic Systems: Uncertainty in weather forecasting, online gaming, AI systems, and network behavior. Events, Outcomes, and Sample Space: Modeling OTP generation and packet transmission. Experimental vs. Theoretical Probability. Conditional Probability and Bayes' Intuition: Spam filtering, disease diagnosis, fraud detection, recommendation systems, and cybersecurity. Randomness in Computer Systems: Random number generation, , and AI decision-making.

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Hands-on Activities

1. Simulate password and OTP security using probability.
2. Develop a simple spam email classifier using Bayes' principle.
3. Analyze network packet loss through probability simulations.
4. Estimate rainfall probability using historical weather data.
5. Design a basic recommendation system using conditional probability.

Module II : Exploring and Understanding Data

Types of Data and Sampling: Social media analytics, healthcare, and surveys. Data Visualization: Dashboards for business intelligence and smart cities. Measures of Central Tendency: Customer behavior, student performance, and app usage analysis. Measures of Dispersion: Quality control, financial risk, and sensor variability. Correlation and Outlier Detection: Fraud detection, anomaly identification, climate analysis, and healthcare monitoring.

Hands-on Activities

1. Analyze smartphone usage or e-commerce datasets.
2. Visualize climate or air-quality data using Python.
3. Detect fraudulent transactions using outlier analysis.
4. Study relationships between variables using correlation.
5. Build an interactive data visualization dashboard.

Module III : Probability Models for Intelligent Systems

Random Variables: Modeling network traffic, sensor readings, and user behavior. Probability Distributions (Binomial, Poisson, Normal, Exponential): Website traffic, packet arrivals, customer queues, manufacturing defects, and cloud computing. Law of Large Numbers and Central Limit Theorem: Data analytics, quality assurance, and AI model evaluation.

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Hands-on Activities

- Simulate website traffic and customer arrivals.
- Analyze Wi-Fi packet transmission using probability distributions.
- Model queue systems for hospitals or banks.
- Visualize different probability distributions using Python.
- Compare simulated and real-world traffic data.

Module IV : Statistical Decision Making

Sampling and Confidence Intervals: Opinion polls, healthcare studies, and product quality assessment. Hypothesis Testing: Software testing and manufacturing decisions. Correlation vs. Causation: Social media trends, business analytics, and scientific studies. Regression and Prediction: Sales forecasting, energy consumption, and academic performance. A/B Testing: Website optimization, digital marketing, and mobile app evaluation.

Hands-on Activities

1. Perform A/B testing on website click-through data.
2. Predict student performance using regression.
3. Compare software versions using hypothesis testing.
4. Analyze customer feedback statistically.
5. Evaluate marketing campaigns using confidence intervals.

Module V : Statistics in AI and Emerging Technologies

Statistical Thinking in AI: Machine learning, predictive analytics, and intelligent decision-making. Bias, Variance, and Uncertainty: AI model evaluation and healthcare diagnostics. Bootstrap and Monte Carlo Simulation: Financial forecasting, engineering reliability, and risk analysis. Anomaly Detection: Cybersecurity, fraud detection, IoT monitoring, and predictive maintenance. Ethical Interpretation of Data: Responsible AI, fairness, privacy, and trustworthy analytics.

Hands-on Activities

1. Detect anomalies in network or IoT sensor data.
2. Simulate financial risk using Monte Carlo methods.
3. Estimate confidence intervals using bootstrap sampling.
4. Evaluate uncertainty in AI prediction results.

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Minor Projects for this semester

1. Smart Spam Email Detection using Bayesian Classification
2. Fraud Detection in Online Transactions
3. Traffic Flow and Queue Simulation
4. Healthcare Risk Prediction
5. Cybersecurity Log Analyzer
6. Air Quality Analytics Dashboard
7. Sports Performance Analytics

Pedagogy

The course adopts an application-driven, experiential, and project-based learning approach to develop statistical thinking and data-driven problem-solving skills. Fundamental concepts of probability and statistics are introduced through real-world case studies from information technology, artificial intelligence, cybersecurity, healthcare, finance, networking, and smart city applications. Learning is reinforced through interactive lectures, inquiry-based discussions, Python-based laboratory sessions, simulations, data visualization, and analysis of real-world datasets using tools such as Jupyter Notebook or Google Colab. Students engage in hands-on activities, collaborative exercises, and case studies to explore uncertainty, interpret data, and support decision-making. This pedagogy emphasizes conceptual understanding, computational thinking, and the practical application of statistical methods in modern IT and Machine Intelligence domains.

GUIDELINES FOR ASSESSMENT AND EVALUATION FOR SEMESTERS V - VIII OF FYUGP (DYD)

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1.2 End-Semester Examination - 60 Marks

This is designed to test comprehensive domain knowledge and applications. The question paper shall have three sections:

- Section A - 20 Marks: Short answer questions to test fundamentals and breadth of domain knowledge.
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- Section C - 20 Marks: This is to be devoted to essay/analytical questions (choice-based) and have to be Open-ended, requiring synthesis, critical thinking, and creativity.

Recommended Textbooks

1. Sheldon M. Ross, *A First Course in Probability*, 10th Edition, Pearson, 2020.
2. Douglas C. Montgomery and George C. Runger, *Applied Statistics and Probability for Engineers*, 8th Edition, Wiley, 2019.
3. Allen B. Downey, *Think Stats: Exploratory Data Analysis in Python*, 2nd Edition, O'Reilly Media, 2014.
4. Trevor Hastie, Robert Tibshirani, and Gareth James, *An Introduction to Statistical Learning with Applications in Python*, Springer, 2023.
5. Jake VanderPlas, *Python Data Science Handbook*, 2nd Edition, O'Reilly Media, 2022.

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Course Code: UMJDTT-813
Credits: 04
Contact Hours: 60

Course Title: Computer Vision
Maximum Marks : 100
Internal Evaluation : 40
External Evaluation : 60

COURSE DESCRIPTION

This course introduces the fundamentals of Computer Vision, enabling computers to understand and interpret images and videos. Students will learn image processing, feature extraction, image classification, object detection, and practical applications using Python and OpenCV through hands-on activities and simple projects.

LEARNING OUTCOMES

1. Understand the basic concepts of computer vision and digital image processing.
2. Apply image preprocessing and feature extraction techniques.
3. Develop simple AI models for image classification and object detection.
4. Build basic computer vision applications using Python and OpenCV.

MODULE I: INTRODUCTION TO COMPUTER VISION

Introduction to computer vision, digital image representation and color models (RGB and Grayscale) for image processing applications, image acquisition techniques for intelligent vision systems, and OpenCV for developing computer vision applications in object detection, image classification, facial recognition, and automated visual inspection.

Hands-On Activities:

1. Read and display images using OpenCV.
2. Perform basic image operations such as resizing, cropping, and color conversion.

MODULE II: IMAGE PREPROCESSING

Image enhancement and filtering for image quality improvement, noise removal techniques for reliable image processing, histogram equalization for contrast enhancement, edge detection and thresholding for image segmentation, and morphological operations for object detection, feature extraction, and computer vision applications.

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Course Title: Computer Vision
Maximum Marks : 100
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Hands-On Activities:

1. Apply image enhancement and filtering techniques.
2. Detect edges and perform thresholding using OpenCV.

MODULE III: FEATURE EXTRACTION

Feature extraction techniques for computer vision applications, corner, edge, shape, and texture feature extraction for object recognition and image analysis, image descriptors for visual representation, and feature matching techniques for image registration, object detection, and image retrieval applications.

Hands-On Activities:

1. Extract image features using OpenCV.
2. Compare different feature extraction methods.

MODULE IV: IMAGE CLASSIFICATION AND OBJECT DETECTION

Introduction to machine learning and deep learning for computer vision applications, Artificial Neural Networks (ANN) and Convolutional Neural Networks (CNN) for image classification and object detection, and performance evaluation and comparative analysis of intelligent vision models.

Hands-On Activities:

1. Develop a simple image classification model.
2. Detect objects in images using OpenCV.

MODULE V: APPLICATIONS OF COMPUTER VISION

Applications of computer vision in face detection, image recognition, medical image analysis, traffic sign recognition, smart surveillance, industrial inspection, and precision agriculture using machine learning and deep learning techniques.

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Hands-On Activities:

1. Build a simple face detection application.
2. Develop a basic image recognition system.

MINOR PROJECTS DURING THE SEMESTER

1. Develop a Smart Attendance System using Face Recognition.
2. Design a Fruit and Vegetable Classification System for Smart Agriculture.
3. Build a Traffic Sign Recognition System for Intelligent Transportation.
4. Develop a Waste Classification System for Smart Waste Management.
5. Design a Defective Product Detection System for Industrial Quality Inspection.

PEDAGOGY

Mentor should introduce each topic through real-world applications such as face recognition, medical imaging, smart surveillance, and autonomous systems to help students understand the importance of computer vision. The course should emphasize experiential learning through Python programming, OpenCV-based image processing, hands-on activities, and mini-projects. Students should be encouraged to solve practical problems using the 'How to Solve It' approach and develop application-oriented skills.

GUIDELINES FOR ASSESSMENT AND EVALUATION FOR SEMESTERS V - VIII OF FYUGP (DYD)

1.Assessment and Evaluation Model

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This is designed to test comprehensive domain knowledge and applications. The question paper shall have three sections:

- Section A - 20 Marks: Short answer questions to test fundamentals and breadth of domain knowledge.
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Recommended Textbooks

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2. David A. Forsyth and Jean Ponce – *Computer Vision: A Modern Approach*, 2nd Edition, Pearson Education, 2011.
3. E. R. Davies – Computer Vision: Principles, Algorithms, Applications, Learning, 5th Edition, Elsevier, 2018.
4. Joseph Howse and Joe Minichino – *Learning OpenCV 4 Computer Vision with Python*, 3rd Edition, Packt Publishing, 2020.
5. Adrian Rosebrock – Practical Python and OpenCV: An Introductory, Example-Driven Guide to Image Processing and Computer Vision, PyImageSearch, 2016.
6. Mohamed Elgendy – *Deep Learning for Vision Systems*, Manning Publications, 2020.

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Course Code: UMJDTT-814

Course Title: IoT Sensor Networks & Data Analytics

Credits: 04
Contact Hours: 60

Maximum Marks : 100
Internal Evaluation : 40
External Evaluation : 60

Course Description:

This course offers a comprehensive and practice-oriented introduction to the Internet of Things (IoT), Wireless Sensor Networks (WSNs), and Data Analytics, emphasizing real-world problem solving and intelligent connected systems. It covers the fundamentals of IoT architecture, embedded devices, sensors, actuators, communication protocols, wireless sensor networks, cloud connectivity, data acquisition, and data analytics for extracting meaningful insights from sensor-generated data. The course focuses on the design and development of smart, connected, and data-driven systems capable of sensing, communicating, analyzing, and responding to real-world environments. Students gain hands-on experience in interfacing sensors with microcontrollers, developing IoT applications, transmitting sensor data to cloud platforms, and applying data analytics techniques for monitoring, prediction, and decision-making.

LEARNING OUTCOMES:

After completing this course, students will be able to:

1. Understand the architecture, components, communication technologies, and applications of IoT and Wireless Sensor Networks.
2. Design and implement IoT systems by integrating sensors, microcontrollers, wireless communication, and cloud platforms for real-world applications.
3. Collect, process, visualize, and analyze sensor-generated data to support monitoring, prediction, and informed decision-making.
4. Develop secure, scalable, and intelligent IoT solutions by integrating cloud computing, data analytics, and AI techniques.

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MODULE I : Building Smart IoT Systems using Sensors and Embedded Devices

Building smart home, smart agriculture, healthcare, industrial and smart city IoT systems. Collecting real-time data using temperature, humidity, motion, GPS, air quality, biomedical and environmental sensors. Developing IoT prototypes using Arduino, ESP32, and Raspberry Pi. Monitoring sensor data using Node-RED, MQTT Explorer, ThingSpeak and Blynk IoT. Visualizing real-time sensor data using Python dashboards.

Hands-On Activities:

1. Identify and interface basic sensors with Arduino or ESP32.
2. Read and display temperature and humidity data using sensors.
3. Build a simple smart lighting system using an LDR and LED.
4. Develop a basic IoT prototype that reads multiple sensor values.

MODULE II : Connecting Smart Devices and Automating IoT Applications

Developing wireless IoT systems using Wi-Fi, Bluetooth, ZigBee, LoRa and MQTT. Building cloud-connected IoT applications using AWS IoT, Azure IoT and Google Cloud IoT. Automating smart homes, smart irrigation and energy management systems. Integrating sensors with cloud databases, mobile applications and web dashboards. Designing IoT workflows for remote monitoring and control.

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Hands-On Activities

1. Connect ESP32 to Wi-Fi and send sensor data to ThingSpeak.
2. Build a real-time IoT dashboard for monitoring sensor values.
3. Control LEDs or appliances remotely using a mobile application.
4. Develop a cloud-connected environmental monitoring system.

MODULE III: Transforming Sensor Data into Intelligent Insights

Developing smart healthcare monitoring systems using ECG, pulse oximeter, body temperature and wearable sensors. Building precision agriculture solutions using soil moisture, humidity, pH, rainfall and weather sensors. Designing smart home automation systems using motion, light, gas and occupancy sensors. Creating industrial monitoring and predictive maintenance systems using vibration, pressure, current and temperature sensors.

Hands-On Activities:

1. Analyze sensor data using Python and Pandas.
2. Create interactive dashboards using Power BI or Google Looker Studio.
3. Visualize real-time IoT data using ThingSpeak or Grafana.
4. Identify trends and anomalies in environmental sensor data.

MODULE IV: IoT Data Analytics for Intelligent Decision Making

Predicting equipment failures and machine health using time-series sensor data, anomaly detection and predictive analytics. Developing energy consumption forecasting systems using smart meter and power sensor data. Designing traffic flow and smart parking analytics using GPS, camera and vehicle sensor data. Creating dashboards for real-time monitoring and decision support using Python, Power BI and Grafana.

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Hands-On Activities:

1. Develop a smart irrigation system based on soil moisture sensing.
2. Build a smart home automation system for lighting and appliance control.
3. Design a smart parking monitoring prototype.
4. Develop a real-time air quality monitoring system with alerts.

MODULE V: Intelligent IoT Systems, Edge AI and Future Applications

Developing Edge AI applications using TinyML, TensorFlow Lite, ESP32 and Raspberry Pi. Building secure IoT systems using device authentication, encryption and secure communication protocols. Designing Digital Twin applications using real-time sensor networks and cloud platforms. Integrating IoT with Artificial Intelligence for autonomous monitoring, predictive maintenance and intelligent automation. Exploring emerging IoT technologies including 5G IoT, Industrial IoT (IIoT), Smart Healthcare, Connected Vehicles and Smart Cities.

Hands-On Activities:

1. Monitor and optimize power consumption of an IoT device.
2. Deploy a complete IoT application with cloud monitoring and dashboards.
3. Present an end-to-end IoT solution addressing a real-world problem.

MINOR PROJECTS DURING THE SEMESTER:

1. Smart Home Automation System using IoT sensors.
2. Smart Agriculture Monitoring and Automated Irrigation System.
3. Air Quality Monitoring and Pollution Analytics Dashboard.
4. Smart Weather Station with Cloud-Based Data Visualization.
5. IoT-Based Energy Consumption Monitoring System.
6. Industrial Equipment Health Monitoring System.
7. Smart Parking Management System using IoT Sensors.
8. Healthcare Patient Monitoring System with Real-Time Alerts.

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PEDAGOGY

The mentor shall introduce each IoT, Sensor Networks, and Data Analytics concept by anchoring it to real-world applications such as smart homes, precision agriculture, industrial automation, environmental monitoring, healthcare systems, intelligent transportation, and smart cities. Concepts will be delivered through progressive laboratory sessions, sensor interfacing exercises, cloud integration, real-time data acquisition, analytics-driven projects, and industrial case studies that closely simulate modern IoT development workflows.

Students will continuously develop connected systems that collect, transmit, analyze, and visualize real-world data while learning system integration, cloud connectivity, cybersecurity, predictive analytics, and intelligent automation. Emphasis will be placed on reliable system design, secure communication, scalability, data-driven decision making, teamwork, documentation, testing, and deployment. The teaching–learning process will adopt a "How to Solve It" problem-solving approach, enabling students to analyze real-world challenges, design intelligent connected systems, and implement industry-ready IoT solutions.

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RECOMMENDED TEXTBOOKS:

1. Arshdeep Bahga and Vijay Madisetti – *Internet of Things: A Hands-on Approach*, Universities Press, 2015.
2. David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Robert Barton, and Jerome Henry – *IoT Fundamentals: Networking Technologies, Protocols, and Use Cases for the Internet of Things*, Cisco Press, 2017.
3. Raj Kamal – *Internet of Things: Architecture and Design Principles*, 3rd Edition, McGraw Hill, 2023.
4. Cuno Pfister – *Getting Started with the Internet of Things*, O'Reilly Media, 2018.
5. Edgar H. Callaway Jr. – *Wireless Sensor Networks: Architectures and Protocols*, CRC Press, 2003.
6. Holger Karl and Andreas Willig – *Protocols and Architectures for Wireless Sensor Networks*, Wiley, 2007.
7. Wes McKinney – *Python for Data Analysis*, 3rd Edition, O'Reilly Media, 2022.
   Joel Grus – *Data Science from Scratch*, 2nd Edition, O'Reilly Media, 2019.
8. Foster Provost and Tom Fawcett – *Data Science for Business*, 2nd Edition, O'Reilly Media, 2023.
9. Jake VanderPlas – *Python Data Science Handbook*, O'Reilly Media, 2023.
10. Thomas Erl, Wajid Khattak, and Paul Buhler – *Big Data Fundamentals*, Prentice Hall, 2016.
11. Andrew Minteer – *Analytics for the Internet of Things (IoT): Intelligent Analytics for Your Intelligent Devices*, Packt Publishing, 2017.

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Four Year Innovative Undergraduate Program
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B. Tech (Information Technology & Mathematical Innovations) (Revised)
Semester VIII
(For the session 2026, 2027, 2028)

Course Code: UMJDTT-815

Course Title: AI Driven Digital Forensics and Cyber threat Intelligence

Credits: 04
Contact Hours: 60

Maximum Marks : 100
Internal Evaluation : 40
External Evaluation : 60

COURSE DESCRIPTION

This course introduces students to the application of Artificial Intelligence in cybersecurity, digital forensics, cyber threat intelligence, and intelligent security operations. Students will learn how intelligent automation are transforming cyber investigations, incident response, malware analysis, threat detection, and security analytics. The course follows a project-based learning approach in which students progressively build AI-assisted cybersecurity applications, digital investigation tools, threat intelligence systems, and intelligent security assistants using modern AI platforms, cybersecurity tools, and real-world datasets. By the end of the course, students will be able to develop AI-enabled cyber solutions while understanding digital evidence, ethical considerations, and responsible AI practices in cybersecurity.

LEARNING OUTCOMES

By the end of this course, students will be able to:

1. Understand the fundamentals of cybersecurity, digital forensics, and cyber threat intelligence.
2. Analyze digital evidence and security data using AI-assisted techniques.
3. Develop AI-powered cybersecurity applications for threat detection, log analysis, and digital investigations.
4. Evaluate AI-enabled cybersecurity solutions while applying ethical, legal, and responsible practices.

MODULE I : Understanding Cyber Threats and Digital Investigations

Understanding how cyber attacks and digital evidence from computers, mobile devices, cloud platforms, IoT devices and enterprise networks. Types of cyber attacks including malware, ransomware, phishing, insider threats and Advanced Persistent Threats (APTs). Digital investigation lifecycle from evidence collection to reporting. Introduction to Python for cybersecurity data analysis and digital investigations.

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Hands-on Activities

1. Investigate a real-world ransomware attack timeline.
2. Analyze phishing emails to identify attack indicators.
3. Explore Windows event logs to identify suspicious activities.
4. Examine network traffic using Wireshark.
5. Prepare a digital forensic investigation report.

MODULE II : AI for Cybersecurity, Digital Forensics and Threat Intelligence

AI for Threat Intelligence and Security Research: Using VirusTotal, MITRE ATT&CK, Shodan, AlienVault OTX for vulnerability analysis, cyber intelligence and attack investigation. AI-assisted Security Operations: Using Splunk, Microsoft Sentinel, Cortex XSOAR and Elastic Security for security monitoring, log analysis and incident response. Prompt Engineering for Cybersecurity: Designing prompts for malware analysis, forensic investigation, vulnerability assessment, threat hunting and security documentation.

Hands-On Activities

1. Analyze sample system and network log files using AI.
2. Build an AI-powered log summarization assistant.
3. Develop a phishing email detection application using AI.
4. Create a simple security monitoring dashboard using cybersecurity datasets.

MODULE III : Building AI-Powered Digital Forensics Applications

Developing malware detection and ransomware analysis systems using malware samples, executable files and behavioral analysis. Building phishing email and malicious URL detection systems using email headers, URLs, Natural Language Processing and Machine Learning. Creating AI-assisted log analysis applications using Windows Event Logs, Linux logs, firewall logs and SIEM data. Building cyber threat intelligence dashboards using Indicators of Compromise (IoCs), CVE databases and threat intelligence feeds.

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Hands-On Activities

1. Analyze sample forensic images or digital evidence using open-source tools.
2. Build a RAG-based cybersecurity knowledge assistant using incident reports.
3. Develop an AI assistant that summarizes forensic investigation reports.
4. Analyze phishing emails, suspicious URLs, or malicious documents using AI.

MODULE IV : Developing Intelligent Cyber Defense Solutions

Building network intrusion detection systems using network packets, NetFlow data, IDS/IPS alerts and firewall logs. Developing endpoint threat detection systems using process monitoring, endpoint telemetry and operating system event logs. Creating cloud security monitoring applications using cloud audit logs, identity events and cloud security telemetry. Developing cyber threat intelligence platforms using MITRE ATT&CK, threat feeds, Indicators of Compromise (IoCs) and vulnerability databases.

Hands-On Activities

1. Build an AI-powered cyber threat intelligence assistant.
2. Develop a multi-agent incident investigation system using CrewAI or AutoGen.
3. Create an AI-based OSINT assistant to collect and summarize cybersecurity information.
4. Build a vulnerability report analyzer that recommends mitigation strategies.

MODULE V : Intelligent Security Operations and Automated Incident Response

Developing AI-powered Security Operations Center (SOC) assistants using Large Language Models, Security Copilot and AI chatbots. Automating threat hunting and incident response workflows using SOAR platforms, APIs and intelligent workflow automation. Deploying secure AI-driven cyber defense solutions using cloud platforms, SIEM dashboards and enterprise security systems. Developing responsible AI solutions for cybersecurity through Explainable AI, cyber ethics, privacy, legal compliance and AI governance.

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Hands-On Activities

1. Evaluate AI-generated security recommendations for accuracy and reliability.
2. Deploy an AI-powered cybersecurity assistant using Streamlit or Gradio.
3. Compare different LLMs for cybersecurity-related tasks.
4. Present a complete AI-enabled cyber investigation solution with technical documentation.

Minor Projects

1. AI-Based Password Strength Analyzer.
2. AI Cybersecurity Awareness Chatbot.
3. AI Phishing Email Detection System.
4. AI Log File Summarization Tool.
5. Cyber Threat Intelligence Dashboard.
6. RAG-Based Cybersecurity Knowledge Assistant.
7. AI Vulnerability Report Analyzer.
8. Digital Forensics Report Generator.

PEDAGOGY

The course will be delivered through a project-based, experiential learning approach that emphasizes learning through practical applications and real-world cybersecurity scenarios. Each module introduces core concepts through case studies, live demonstrations, guided laboratory sessions, and hands-on development using AI-enabled cybersecurity tools and platforms. Students will progressively build AI-assisted cybersecurity applications, digital forensic investigation tools, cyber threat intelligence systems, and intelligent security assistants methodology. Laboratory sessions will integrate cybersecurity datasets, digital evidence, system logs, network traffic, threat intelligence feeds, and open-source investigation tools with modern AI technologies, enabling students to automate analysis, support investigations, and develop intelligent cybersecurity solutions.

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GUIDELINES FOR ASSESSMENT AND EVALUATION FOR SEMESTERS V - VIII OF FYUGP (DYD)

1.Assessment and Evaluation Model

Features of this Model of Assessment and Evaluation are:

- Domain knowledge: Tested through class tests and end-sem examination.
- Creativity: Assignments, seminars, open-ended examination questions.
- Hands-on experience: Practical tasks and project work to build real-world problem-solving skills.

Following scheme of assessment and evaluation shall be followed for the last two years of FYUGP (DYD):

1.1 Continuous Internal Assessment (CIA) - 40 Marks

This ensures sustained engagement and assessment beyond the final exam. CIA shall have the following components:

- Class Tests and Quizzes (10 Marks): This shall focus on concepts clarity, analytical ability comprising 2 or 3 short written or online tests.
- Assignments and Creative Tasks (10 Marks): This shall be focused on Problem-solving or reflective assignments and Open-ended questions to assess creativity.
- Practical and Hands-on Activities (15 Marks): This shall have the activities like Lab work, field tasks, coding assignments, design exercises, case study etc.
- Seminar and Presentation (5 Marks): This shall have individual or group presentation to be evaluated for communication, originality, creativity, and depth.

1.2 End-Semester Examination - 60 Marks

This is designed to test comprehensive domain knowledge and applications. The question paper shall have three sections:

- Section A - 20 Marks: Short answer questions to test fundamentals and breadth of domain knowledge.

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- Section B - 20 Marks: Problem-solving
-
- and applications of the domain knowledge. The problems need to be conceptual numerical and case-based related to real world applications.

- Section C - 20 Marks: This is to be devoted to essay/analytical questions (choice-based) and have to be Open-ended, requiring synthesis, critical thinking, and creativity.

RECOMMENDED TEXTBOOKS

1. Nina Godbole and Sunit Belapure, *Cyber Security: Understanding Cyber Crimes, Computer Forensics and Legal Perspectives*, Wiley India, Latest Edition.
2. Bill Nelson, Amelia Phillips, Christopher Steuart, *Guide to Computer Forensics and Investigations*, Cengage Learning, Latest Edition.
3. Jason Andress, *The Basics of Information Security: Understanding the Fundamentals of InfoSec in Theory and Practice*, Elsevier, Latest Edition.
4. Richard Bejtlich, *The Practice of Network Security Monitoring: Understanding Incident Detection and Response*, No Starch Press.
5. Chris Sanders and Jason Smith, *Applied Network Security Monitoring*, Syngress.
6. Sammie Bae, *Digital Forensics with Kali Linux*, Packt Publishing.
7. Michael Sikorski and Andrew Honig, *Practical Malware Analysis*, No Starch Press.

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Course Code: UMJDTP-816	Course Title : Project 1
Credits: 06	Maximum Marks : 150
	Internal Evaluation : 100
	External Evaluation : 50

Guidelines for Interdisciplinary Major Projects

To promote transdisciplinarity, experiential learning, skill development and local to global relevance the essence of the DYD program the BOS of SIIEDC resolves to have at least two semester long major projects carrying 150 marks each in semesters V to VIII.

1. Objectives

a) Integrate theoretical knowledge across disciplines.
b) Encourage research-oriented and inquiry-driven learning.
c) Develop analytical, digital, and communication competencies.
d) local, regional, or global developmental challenges.
e) Enhance employability and innovation skills.

2. Nature and Scope of Projects

Projects may include interdisciplinary academic research, mathematical or computational modelling, design thinking and technological solutions, community engagement initiatives, policy analysis of educational innovation and entrepreneurship or sustainability-focused projects. Each project must clearly demonstrate integration of at least two courses of the current semester. Transdisciplinary project in collaboration are also encouraged.

3. Structure and Supervision

A group of 3-5 students shall work on the project where interdisciplinary team formation is encouraged. One Primary Mentors shall guide the project at least one from another discipline(s)/course(s) may be appointed where required. The project shall be conducted over one semester with the following suggested indicators:

a) Proposal Development and Approval
b) Literature Review and Conceptual Framework
c) Methodology Design
d) Implementation/Analysis
e) Mid-Term Review
f) Final Report Submission
g) Presentation and Viva-Voce

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Course Code: UMJDTP-816
Credits: 06

Course Title : Project 1
Maximum Marks : 150
Internal Evaluation : 100
External Evaluation : 50

4. Proposal Guidelines

Each project proposal shall include

a) Title and thematic area.
b) Courses/disciplines integrated.
c) Problem statement and motivation.
d) Objectives and expected outcomes.
e) Brief literature review.
f) Methodology and tools.
g) Timeline with milestones,
h) Ethical considerations (if applicable).

Approval shall be granted by the DYD Academic/Project Committee.

5. Evaluation Scheme (Credits-6) Marks: 150

a)Continuous Assessment - 40 Marks

Proposal quality and originality	10
Literature review	10
Methodology planning	10
Progress, logbook & mentor assessment	10

b)Final Report - 60 Marks

Problem formulation and clarity	10
Interdisciplinary integration	10
Methodology and analysis	20
Innovation and societal/technological relevance	10
Writing quality and references	10

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Credits: 06

Course Title : Project 1
Maximum Marks : 150
Internal Evaluation : 100
External Evaluation : 50

c)Presentation & Viva Voce-50 Marks

Conceptual understanding	15
Implementation/results/outcome	15
Communication skills	5
Response to questions	10
Collaboration/team work	5

Students shall submit a hardbound copy of the project report, along with a soft copy containing the PDF of the report and all relevant datasets, source code, and design artefacts. In addition, students mast submit the presentation slides prepared for the project evaluation and a duly maintained project diary/logbook certified by the mentor(s).

6. Academic Integrity and Ethics

To meet academic integrity and professional ethics, proper citation and avoidance of plagiarism are mandatory. Also responsible use of AI tools must be declared.

7. Role of the DYD Project Committee

The committee shall approve project proposals, ensure interdisciplinary balance, monitor the progress. standardize evaluation procedures and maintain a repository of DYD projects.

8. Timeline of the Project

The project work is required to be initiated in the following phases:
Week 1-3: Proposal preparation and approval
Week 4-6: Literature review, data collection and methodology
Week 8-12: Implementation/Analysis
Week 13: Final submission and viva-voce.

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Credits: 06	Maximum Marks : 150
	Internal Evaluation : 100
	External Evaluation : 50

9. Outcomes

By the end of 8th semester, the students shall have atleast any one of the following outcomes:

a) Application for Intellectual Property Rights (IPR) such as design patent, copyright, etc.
b) Submission of a high-quality research article in a Scopus/SCI/SCIE indexed journal, with the manuscript accepted for review/publication.
c) Conversion of the idea into a start-up, including submission of an application to an incubator for start-up support.

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Course Code: UMJDTP-817	Course Title : Project 2
Credits: 06	Maximum Marks : 150
	Internal Evaluation : 100
	External Evaluation : 50

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To promote transdisciplinarity, experiential learning, skill development and local to global relevance the essence of the DYD program the BOS of SIIEDC resolves to have at least two semester long major projects carrying 150 marks each in semesters V to VIII.

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d) local, regional, or global developmental challenges.
e) Enhance employability and innovation skills.

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Projects may include interdisciplinary academic research, mathematical or computational modelling, design thinking and technological solutions, community engagement initiatives, policy analysis of educational innovation and entrepreneurship or sustainability-focused projects. Each project must clearly demonstrate integration of at least two courses of the current semester. Transdisciplinary project in collaboration are also encouraged.

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A group of 3-5 students shall work on the project where interdisciplinary team formation is encouraged. One Primary Mentors shall guide the project at least one from another discipline(s)/course(s) may be appointed where required. The project shall be conducted over one semester with the following suggested indicators:

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Course Code: UMJDTP-817
Credits: 06

Course Title : Project 2
Maximum Marks : 150
Internal Evaluation : 100
External Evaluation : 50

4. Proposal Guidelines

Each project proposal shall include

a) Title and thematic area.
b) Courses/disciplines integrated.
c) Problem statement and motivation.
d) Objectives and expected outcomes.
e) Brief literature review.
f) Methodology and tools.
g) Timeline with milestones,
h) Ethical considerations (if applicable).

Approval shall be granted by the DYD Academic/Project Committee.

5. Evaluation Scheme (Credits-6) Marks: 150

a)Continuous Assessment - 40 Marks

Proposal quality and originality	10
Literature review	10
Methodology planning	10
Progress, logbook & mentor assessment	10

b)Final Report - 60 Marks

Problem formulation and clarity	10
Interdisciplinary integration	10
Methodology and analysis	20
Innovation and societal/technological relevance	10
Writing quality and references	10

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Credits: 06

Course Title : Project 2
Maximum Marks : 150
Internal Evaluation : 100
External Evaluation : 50

c) Presentation & Viva Voce-50 Marks

Conceptual understanding	15
Implementation/results/outcome	15
Communication skills	5
Response to questions	10
Collaboration/team work	5

Students shall submit a hardbound copy of the project report, along with a soft copy containing the PDF of the report and all relevant datasets, source code, and design artefacts. In addition, students mast submit the presentation slides prepared for the project evaluation and a duly maintained project diary/logbook certified by the mentor(s).

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Credits: 06	Maximum Marks	: 150
	Internal Evaluation	: 100
	External Evaluation	: 50

9. Outcomes

By the end of 8th semester, the students shall have atleast any one of the following outcomes:

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b) Submission of a high-quality research article in a Scopus/SCI/SCIE indexed journal, with the manuscript accepted for review/publication.
c) Conversion of the idea into a start-up, including submission of an application to an incubator for start-up support.